PSYC 4 Midterm 2 Notes（Ch 6 to 10）

作答規則
1. 先讀最後一句問句，遮住選項，心裡說答案。
2. 看到一樣的就圈起來，其他選項每個找一個錯字才劃掉。
3. 題目說 normal、intact 的，先劃掉相關選項。
4. **選項裡有下面的成對概念，就算確定也標？，翻這頁核對方向。**
5. 第一個答案留著，只有在題目裡找到字證明它錯才改。

Rods 桿細胞 周邊、暗光敏感、看不到顏色、多個接一個雙極細胞（模糊）	**Cones 錐細胞** 中央小窩、要亮光、看顏色、一對一（清楚）
Photoreceptors in the DARK 黑暗 去極化、cGMP 高、鈉通道開、一直放 glutamate、壓住雙極細胞	**Photoreceptors in LIGHT 有光** cGMP 被分解、鈉通道關、過極化、glutamate 變少、雙極細胞被放開
Wavelength 波長 顏色 hue（光）／音高 pitch（聲音＝frequency）	**Amplitude 振幅** 亮度 brightness（光）／響度 loudness（聲音）
Iris 虹膜 肌肉，窗簾，控制瞳孔大小	**Pupil 瞳孔** 只是一個洞，什麼都不控制
Fovea 中央小窩 錐細胞最多，看得最清楚	**Optic disc 視神經盤** 軸突離開的後門，沒有感光細胞＝盲點
Trichromatic 三色說 錐細胞那一層（三種錐細胞混色）	**Opponent-process 拮抗說** 神經節細胞那一層（紅綠、藍黃、黑白；後像）
Ventral stream 腹側 顳葉，WHAT：認東西、認臉（壞了＝臉盲 prosopagnosia）	**Dorsal stream 背側** 頂葉，WHERE／動態：MT 壞了＝看不到動 akinetopsia
Simple cell 固定角度＋固定位置	**Complex cell** 固定角度、位置不限，喜歡會動的
Base 基底膜底部 窄、硬、靠近 oval window：高音；噪音先打壞這裡	**Apex 頂端** 寬、軟：低音
Touch 觸覺路線 同側往上，在延腦 medulla 交叉	**Pain 痛覺路線** 一進脊髓馬上交叉
脊髓單側受傷 傷哪邊，哪邊沒觸覺	 痛覺沒的是另一邊
Mechanoreceptors 壓力、震動、拉長（Pacinian：壓洋蔥、鈉進去）	**Nociceptors** 痛：傷害、極端溫度、化學物質
Gate：短暫緊急 把痛的門關上（受傷沒感覺）	**Gate：長期壓力、缺睡、餓** 門打開（更痛）
Smell 嗅覺 跳過視丘，直通杏仁核、海馬迴	**Taste 味覺** 延腦 → 視丘 → 味覺皮質

Muscle spindle 肌梭 偵測拉長 length；膝跳反射，單突觸	**Golgi tendon organ** 偵測張力 tension；抑制性中間神經元，讓肌肉放鬆
Monosynaptic 單突觸 膝跳反射，最快	**Polysynaptic 多突觸** 縮手反射（痛覺受器＋中間神經元）
Dorsal 背側 感覺進來 IN	**Ventral 腹側** 運動出去 OUT（alpha 運動神經元在腹角）
Corticospinal 交叉點 延腦和脊髓交界處	**終點** 脊髓腹角（接 alpha 運動神經元）
Acetylcholine 神經肌肉接合處 → 鈉進肌肉 → 收縮	**Dopamine** 基底核油門（黑質），也管快樂
Gas 油門＝substantia nigra 多巴胺不夠＝帕金森：靜止會抖、很難開始動、先憂鬱、先單側	**Brakes 煞車＝striatum** 壞了＝杭丁頓：停不下來的抽動、顯性基因
FES 讀出訊號 OUT，繞過脊髓，刺激肌肉	**DBS** 送電刺激 IN 到大腦（帕金森）
Drive reduction 驅力降低 推 push：補足缺乏（餓了吃餅乾）	**Reward 獎賞** 拉 pull：誘因（飽了還吃蛋糕）
Homeostasis 概念、過程	**Set point** 具體的設定值（例如 37 度）
Endotherm 內溫 自己產熱（松鼠、人）	**Ectotherm 外溫** 靠環境（蜥蜴曬太陽）
Hyperthermia 體溫太高（hyper 太多）	**Hypothermia** 體溫太低（hypo 太少）
Hypotonic 低張 外面稀 → 水跑進細胞 → 脹大	**Hypertonic 高張** 外面濃（吃鹹、3% 鹽水）→ 水跑出來 → 縮水
Osmotic thirst 滲透壓口渴 吃鹹，細胞縮水，OVLT 偵測；補水就好	**Hypovolemic thirst 低血容量口渴** 失血、拉肚子，baroreceptors、腎臟、angiotensin II → subfornical organ；要補水＋鹽
Insulin 胰島素 血糖高時：葡萄糖進細胞、存成肝醣	**Glucagon 升糖素** 血糖低時：肝醣變回葡萄糖
Ghrelin 空腹的胃 → 弓狀核 → orexin → 想吃	**Leptin** 脂肪說「存夠了」；胖的人常有阻抗
Agonist 促效劑 讓神經傳導物質作用變強（古柯鹼擋回收）	**Antagonist 拮抗劑** 擋住受體，作用變弱
MIH 殺掉 Müllerian（女性管道）	**Testosterone** 養 Wolffian（男性管道）
DHT（要 5-alpha reductase） 外生殖器：陰莖	**Estradiol（要 aromatase）** 大腦男性化
Organizational 塑造 出生前荷爾蒙，永久改變大腦（CAH）	**Activational 啟動** 青春期以後荷爾蒙，啟動行為
外來睪固酮 負回饋：壓低 GnRH、LH、FSH → 睪丸萎縮	 不是調高

Study Guide 單字表：情境裡看到這些，答案就是它

第 6 章		
Sensation sen-SAY-shun	感覺：收訊號 receiving signals and turning them into neural signals	**eyes or ears test normal; signals are received** 眼睛、耳朵正常、收得到
Perception per-SEP-shun	知覺：大腦詮釋 the brain organizing and interpreting signals	**receives signals but cannot recognize; two people see the same image differently** 收得到但認不出、兩個人看同一張圖看法不同
Bottom-up processing BOT-um UP	由下而上 building meaning from small details	**first time seeing it; no expectations; reading letter by letter** 第一次看到、沒有預期、一個字母一個字母讀
Top-down processing TOP DOWN	由上而下 using knowledge and expectations	**expectations, familiarity, missing a typo, stereotypes** 期待、熟悉、沒看到錯字、刻板印象
Photon FOH-ton	光子 a particle of light	**more photons = brighter** 光子變多＝更亮
Wavelength WAVE-length	波長 distance between waves; codes color	**color; red vs green light; 530 nm** 顏色、紅光綠光、530 nm
Amplitude AM-plih-tood	振幅 height of the wave; codes brightness or loudness	**brighter, dimmer, louder, taller wave** 更亮、更暗、更大聲、波變高
Refraction rih-FRAK-shun	折射 bending of light	**blurry underwater; goggles; focusing at the cornea** 水裡看不清、蛙鏡、角膜、對焦
Cornea KOR-nee-uh	角膜 clear front surface; does most bending	**where light enters from the air** 光從空氣進眼睛那一層
Lens LENZ	水晶體 adjusts shape to focus near and far	**thickens to see near objects; fails with age** 近看要變厚、老花
Iris EYE-ris	虹膜 colored muscle that sets pupil size	**pupil gets bigger or smaller; eye drops** 瞳孔變大變小、眼藥水
Pupil PYOO-pil	瞳孔 the opening; controls nothing	**just an opening** 只是洞
Fovea FOH-vee-uh	中央小窩 center of retina, packed with cones	**looking straight at it; sharpest; color** 正對著看、最清楚、顏色

Optic disc OP-tik DISK	視神經盤 where axons leave; no receptors	**blind spot; a dot disappears** 盲點、點消失
Rods RODZ	桿細胞 sensitive, dim light, no color, periphery	**dim light; corner of the eye; look to the side at a faint star** 暗處、眼角、星星往旁邊看
Cones KOHNZ	錐細胞 bright light, color, detail, fovea	**color, detail, needs bright light** 顏色、細節、要亮
Bipolar cells by-POH-ler	雙極細胞 middle layer; released when light arrives	**held down in the dark; fire when light arrives** 黑暗中被壓住、光來才發
Ganglion cells GANG-glee-on	神經節細胞 last layer; first to fire action potentials; axons form optic nerve	**last layer; opponent pairs; leave through the optic disc** 最後一層、拮抗配對、走出視神經盤
Horizontal / amacrine cells hor-ih-ZON-tul / AM-uh-krin	水平／無軸突細胞 side connections; lateral inhibition	**side connections; sharper edges** 側向連接、邊緣變清楚
Glutamate (in the dark) GLOO-tuh-mate	麩胺酸 released by photoreceptors in the dark to inhibit bipolar cells	**released all the time in the dark; less in light** 黑暗中一直放、光來變少
cGMP see-jee-em-PEE	cGMP keeps sodium channels open in the dark; light breaks it down	**not broken down = acts like darkness** 不被分解＝一直像黑暗
Receptive field (center-surround) ree-SEP-tiv FEELD	感受野 area a cell responds to; center excites, surround inhibits	**center light strong; surround light weak; whole field cancels** 照中心強、照外圈弱、整片抵消
Lateral inhibition LAT-er-al in-hih-BISH-un	側向抑制 neighbors inhibit each other to sharpen edges	**edges, contrast, Hermann grid** 邊緣、對比、Hermann grid
Trichromatic theory try-kroh-MAT-ik	三色說 three cone types mix colors	**cones; color mixing; red + green = yellow** 錐細胞、混色、紅＋綠＝黃
Opponent-process theory oh-POH-nent	拮抗說 red-green, blue-yellow pairs in ganglion cells	**afterimage; stare, then see the opposite color** 後像、盯久了看到相反色
Simple / complex cells SIM-pul / KOM-plex	簡單／複雜細胞 V1 cells: fixed spot vs anywhere, likes motion	**fixed spot vs anywhere and moving** 固定位置 vs 哪裡都行會動
Ventral stream VEN-trul	腹側路線 to temporal lobe: WHAT (objects, faces); damage = prosopagnosia (PROS-oh-pag-NOH-zhuh), face blindness	**recognize objects and faces; face blindness** 認東西、認臉、臉盲

Dorsal stream DOR-sul	背側路線 to parietal lobe: WHERE and motion (area MT); damage = akinetopsia (ay-kin-eh-TOP-see-uh), motion blindness	**cannot see motion; frozen snapshots** 看不到動、像照片一格格
第 7 章		
Frequency FREE-kwen-see	頻率 cycles per second; codes pitch	**pitch; cycles per second** 音高、每秒幾次
Basilar membrane BAZ-ih-ler	基底膜 in cochlea; base = high pitch, apex = low pitch	**cannot hear high pitch = base damaged** 高音聽不到＝底部壞
Hair cells / stereocilia steer-ee-oh-SIL-ee-uh	毛細胞／靜纖毛 mechanoreceptors; bending pulls channels open	**bending pulls channels open mechanically, not voltage** 毛彎曲、機械性拉開、不是電壓
Medial geniculate nucleus (MGN) jeh-NIK-yoo-lit	內側膝狀核 thalamus relay for hearing	**hearing relay in the thalamus** 聽覺經視丘
Auditory cortex AW-dih-tor-ee	聽覺皮質 temporal lobe	**temporal lobe** 顳葉
McGurk effect mik-GURK	McGurk 效應 seeing lips changes what you hear	**sound ba + lips ga = hear da; eyes closed restores ba** 聲音 ba 嘴型 ga 聽到 da、閉眼恢復
Association cortex uh-soh-see-AY-shun	聯合皮質 areas that combine senses	**combines vision with hearing or touch** 整合視覺和聽覺／觸覺
Somatosensory cortex soh-mat-oh-SEN-sor-ee	體感皮質 postcentral gyrus; receives touch	**tingling; feeling; behind the central sulcus** 刺刺的、感覺、中央溝後面
Motor cortex MOH-ter	運動皮質 precentral gyrus; sends movement commands	**twitching; sending commands; in front of the central sulcus** 抽動、下命令、中央溝前面
Mechanoreceptors meh-KAN-oh-ree-sep-terz	機械受器 respond to pressure, vibration, stretch	**pressure, vibration, Pacinian, hair cells** 壓、震動、Pacinian、毛細胞
Pacinian corpuscle pah-SIN-ee-un KOR-pus-ul	Pacinian 小體 onion-shaped; sudden pressure and vibration	**responds to sudden pressure, ignores steady pressure** 突然壓才有反應、持續壓不理
Thermoreceptors THER-moh-ree-sep-terz	溫度受器 respond to warm or cold	**warm or cold (not pain)** 冷熱（不是痛）
Nociceptors NOH-sih-sep-terz	痛覺受器 respond to damage, extreme heat, chemicals	**injuries not felt; normal axons and myelin but no pain** 受傷沒感覺、軸突髓鞘正常卻不痛

Proprioceptors proh-pree-oh-SEP-terz	本體感受器 in muscles; length and tension	**knows hand position with eyes closed; cannot tell where feet are without looking** 閉眼知道手在哪、不看不知道腳在哪
Vestibular sense ves-TIB-yoo-ler	前庭覺 balance; fluid in semicircular canals	**dizzy after spinning; balance** 轉圈頭暈、平衡
Two-point discrimination dis-krim-ih-NAY-shun	兩點辨識 telling two touches apart	**two points on fingertip, one on upper arm** 指尖兩點、上臂一點
Dermatome DER-muh-tome	皮節 skin area served by one spinal level	**no feeling below a certain level** 某個高度以下沒感覺
Dorsal column pathway DOR-sul KOL-um	觸覺路線 touch; same side, crosses in medulla	**touch lost on the same side** 同側失去觸覺
Spinothalamic pathway spy-noh-thuh-LAM-ik	痛覺路線 pain; crosses right away in cord	**pain lost on the opposite side** 對側失去痛覺
Homunculus hoh-MUNK-yoo-lus	小人圖 body map; hands, lips large; face next to hand	**area depends on fine control; touching the face feels like the missing hand** 面積看精細度、摸臉覺得手被碰
Plasticity plas-TIS-ih-tee	可塑性 map changes with use; slow, not instant	**violinist; weeks after amputation (not seconds)** 小提琴家、截肢後幾週（不是幾秒）
Phantom limb pain FAN-tum LIM	幻肢痛 pain in a missing limb	**amputee; still clenching** 截肢、還在握拳
Mirror box MIR-er BOX	鏡箱 vision supplies the missing feedback	**seeing the hand open releases the cramp** 看到手張開就鬆開
Gate theory GATE	閘門理論 touch input closes a spinal gate on pain	**rubbing helps; stress and sleep loss make it worse** 揉一揉、壓力缺睡會更痛
Olfactory bulb ol-FAK-tor-ee	嗅球 smell; skips thalamus, to amygdala and hippocampus	**a smell brings back old memories** 聞到就想起往事
Gustatory pathway GUS-tuh-tor-ee	味覺路線 taste: medulla, thalamus, gustatory cortex	**taste goes through the thalamus** 味覺經視丘
Insula IN-suh-luh	腦島 where taste, smell, touch combine into flavor	**pinched nose leaves only sweet** 捏鼻子只剩甜
第 8 章		
Muscle spindle SPIN-dul	肌梭 detects stretch (length)	**knee jerk; muscle stretched** 膝跳、被拉長

Golgi tendon organ GOAL-jee TEN-dun	高爾基腱器 detects tension (force); relaxes muscle	**drops a heavy object; too much force makes the muscle relax** 太重放手、用力太大鬆開
Alpha motor neuron AL-fuh	alpha 運動神經元 in ventral horn; makes muscle contract	**ventral horn; last nerve to the muscle** 腹角、最後一條神經
Acetylcholine (ACh) uh-see-til-KOH-leen	乙醯膽鹼 neuromuscular junction transmitter	**muscles do not contract although the nerves are normal** 肌肉不收縮但神經正常
Decussation dek-uh-SAY-shun	交叉 crossing; corticospinal at medulla-cord junction	**asks where the axons cross** 問在哪裡 cross
Monosynaptic reflex mon-oh-sih-NAP-tik	單突觸反射 one synapse; knee jerk	**tapping the knee** 敲膝蓋
Polysynaptic reflex pol-ee-sih-NAP-tik	多突觸反射 with interneurons; withdrawal	**touching something hot and pulling away** 碰燙縮手
Reciprocal inhibition rih-SIP-roh-kul	相互抑制 opposite muscle relaxes	**biceps contracts, triceps relaxes** 二頭肌收、三頭肌鬆
Somatotopic map soh-mat-oh-TOP-ik	體部位地圖 body map in motor cortex	**fingers and lips take a lot of space** 手指嘴唇佔很多
FES EF-EE-ES	功能性電刺激 reads cortex signals out, stimulates muscles, bypasses cord	**paralyzed patient uses an implant to grab a cup** 四肢癱瘓靠晶片拿杯子
Cerebellum sair-uh-BEL-um	小腦 compares intended vs actual; corrects errors	**overshoots, oscillates, cannot adapt to prisms, sobriety test** 伸過頭、來回晃、稜鏡適應不了、酒測
Substantia nigra sub-STAN-shuh NY-gruh	黑質 dopamine; the GAS	**Parkinson; depression comes first** 帕金森、先憂鬱
Striatum (caudate, putamen) stry-AY-tum	紋狀體 the BRAKES	**Huntington; caudate and putamen shrink** 杭丁頓、尾核殼核萎縮
Parkinson disease PAR-kin-sun	帕金森 not enough gas: rest tremor, hard to start moving	**rest tremor; freezes in doorways** 靜止會抖、門口卡住
Huntington disease HUN-ting-tun	杭丁頓 broken brakes: jerky movements, dominant gene	**jerky movements that will not stop; a parent had it** 停不下來的抽動、父母也有
Demyelination dee-my-uh-lih-NAY-shun	髓鞘流失 myelin lost; slow, scattered signals	**cortex and muscle normal; signals slow and out of step** 皮質肌肉正常、訊號慢又散

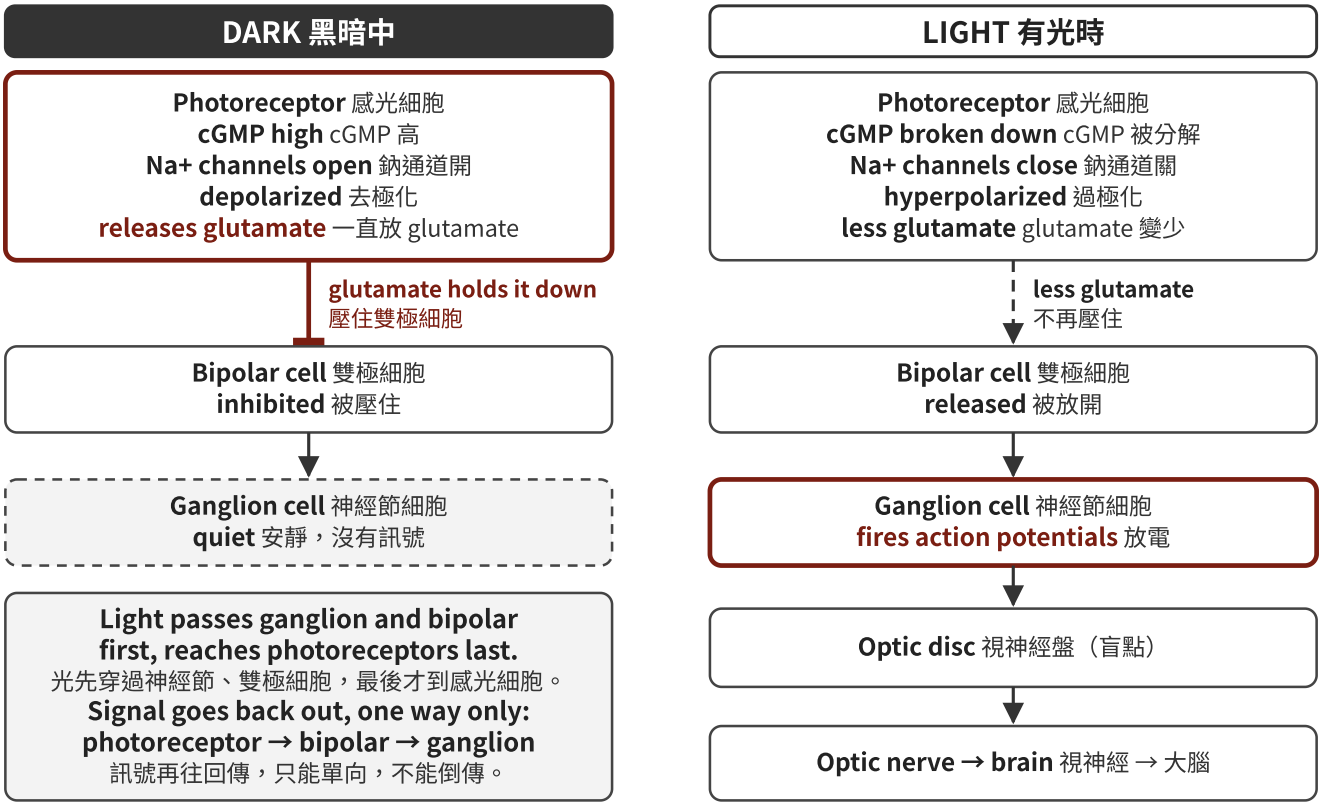
Mirror neurons MIR-er NOOR-onz	鏡像神經元 fire when doing and watching	**watching someone else act; imitation** 坐著看別人做、模仿
第 9 章		
Drive reduction DRIVE rih-DUK-shun	驅力降低 push: removing a need	**hungry, eats until not hungry** 餓了吃到不餓
Reward / incentive rih-WARD / in-SEN-tiv	獎賞 pull: wanting something pleasant	**full but still orders cake** 飽了還吃蛋糕
Homeostasis hoh-mee-oh-STAY-sis	恆定 the concept of keeping balance	**keeping things stable (the concept)** 維持穩定（概念）
Set point SET POINT	設定值 the specific target value	**37 degrees; raised in fever** 37 度、發燒調高
Endotherm EN-doh-therm	內溫動物 makes own heat	**squirrel, human** 松鼠、人
Ectotherm EK-toh-therm	外溫動物 heat from environment	**lizard basking in the sun** 蜥蜴曬太陽
Hyperthermia / hypothermia hy-per-THER-mee-uh / hy-poh-	體溫過高／過低 too hot / too cold	**hyper = too much, hypo = too little** hyper 多、hypo 少
Preoptic area (POA) pree-OP-tik	視前區 hypothalamic warm/cold sensing neurons	**fever, shivering, sweating** 發燒、發抖、流汗
Pyrogens PY-roh-jenz	致熱原 raise the set point (fever)	**normal temperature but shivering** 正常體溫卻發抖
Isotonic eye-soh-TON-ik	等張 equal concentration; no net water movement	**no change** 不變
Hypotonic hy-poh-TON-ik	低張 outside more dilute; water enters; cell swells	**very dilute IV; drinking too much water** 很稀的點滴、喝太多水
Hypertonic hy-per-TON-ik	高張 outside more concentrated; water leaves; cell shrinks	**salty food; 3% saline** 吃鹹、3% 鹽水
Osmotic thirst oz-MOT-ik	滲透壓口渴 cells shrink; OVLT osmoreceptors	**salty food; no blood loss** 吃鹹、沒失血
Hypovolemic thirst hy-poh-voh-LEE-mik	低血容量口渴 blood volume drops; baroreceptors, angiotensin II to subfornical organ	**blood loss, diarrhea; needs salt too** 失血、拉肚子；要補鹽
ADH (vasopressin) AY-DEE-AYCH	抗利尿激素 kidneys save water	**saves water; less urine** 省水、尿變少

Glucoreceptors gloo-koh-ree-SEP-terz	葡萄糖受器 detect blood glucose	**detect blood glucose** 偵測血糖
Insulin IN-suh-lin	胰島素 high sugar: store it	**right after a meal** 剛吃飽
Glucagon GLOO-kuh-gon	升糖素 low sugar: release it from glycogen	**long fasting; marathon** 餓很久、馬拉松
Ghrelin GREL-in	飢餓素 stomach hunger signal	**empty stomach; wants to eat** 空腹、想吃
Leptin LEP-tin	瘦素 fat says stores are full; resistance in obesity	**obese person with high leptin is still hungry** 胖的人 leptin 高還是餓
Orexins oh-REK-sinz	食慾素 lateral hypothalamus; promote eating	**start eating** 啟動進食
Arcuate nucleus AR-kyoo-it	弓狀核 hypothalamus hub for hunger signals	**where ghrelin and leptin act** ghrelin、leptin 作用的地方
VTA and nucleus accumbens NOO-klee-us uh-KUM-benz	獎賞迴路 dopamine reward circuit	**pleasure, addiction** 快樂、成癮
Cocaine koh-KANE	古柯鹼 blocks dopamine reuptake: agonist	**blocks the transporter = stronger effect** 擋轉運體＝作用變強
第 10 章		
SRY / TDF ES-AR-WHY	SRY／睪丸決定因子 Y gene that makes testes	**the first domino** 第一張骨牌
Wolffian ducts WOOL-fee-un	沃爾夫管 male ducts; need testosterone	**grown by testosterone** 睪固酮養
Müllerian ducts myoo-LAIR-ee-un	繆勒管 female ducts; default	**killed by MIH** MIH 殺掉
MIH EM-EYE-AYCH	繆勒管抑制激素（課本叫 AMH） from testes; kills Müllerian (textbook calls it AMH)	**present whenever there are testes** 有睪丸就有
DHT / 5-alpha reductase DEE-AYCH-TEE	雙氫睪固酮 makes penis	**missing = no penis, otherwise male** 缺了＝沒陰莖、其他男性
Aromatase uh-ROH-muh-tase	芳香酶 turns testosterone into estradiol in neurons	**missing = female brain** 沒有＝大腦女性
Alpha-fetoprotein AL-fuh fee-toh-PROH-teen	甲型胎兒蛋白 blocks mother's estrogen from fetal brain	**why the mother's estrogen has no effect** 媽媽雌激素為什麼不影響
Sexual dimorphism dy-MOR-fizm	性別雙型 structures differing by sex	**INAH-3, SDN** INAH-3、SDN

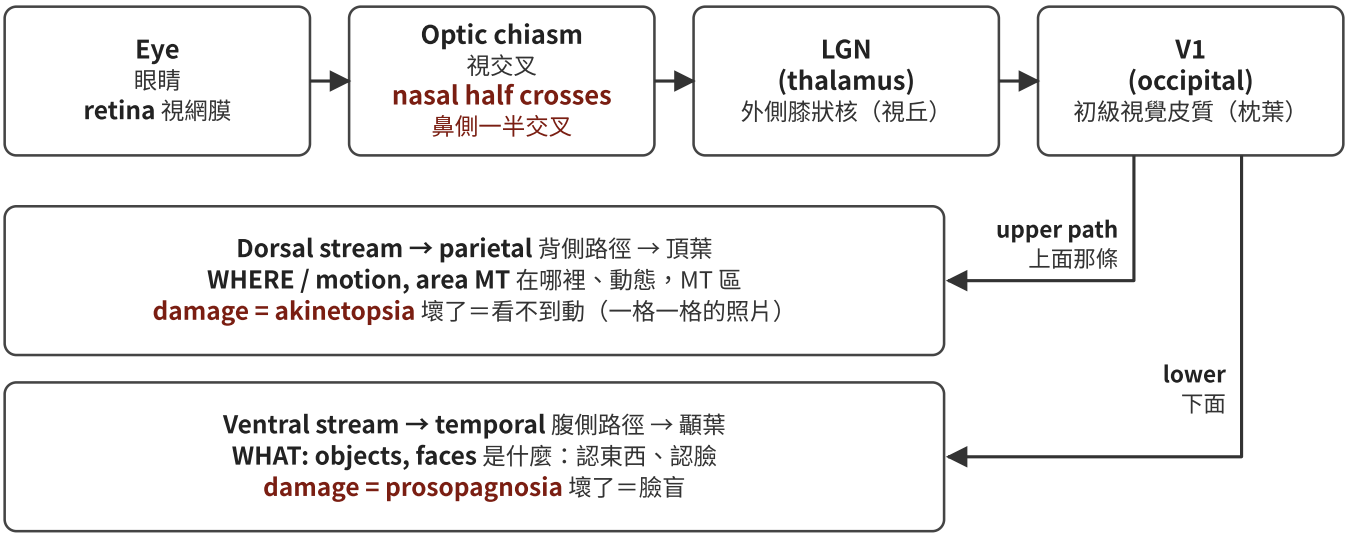
AIS AY-EYE-ES	雄性素不敏感 receptors don't respond to androgens	**XY with female appearance and no ducts** XY 外表女性、沒管道
PMDS PEE-EM-DEE-ES	持續繆勒管症候群 no working MIH	**both duct systems** 兩套管道都有
Klinefelter (XXY) KLINE-fel-ter	克氏症 male, low testosterone	**tall, small testes** 高個子、睪丸小
Turner (XO) TUR-ner	透納氏症 female, missing one X	**short, no puberty** 矮、沒有青春期
CAH SEE-AY-AYCH	先天性腎上腺增生 high prenatal androgens in XX	**girl prefers rough play** 女孩喜歡粗野遊戲
Organizational effects or-guh-nih-ZAY-shun-ul	塑造作用 prenatal, permanent brain shaping	**before birth; shows up in childhood** 出生前、童年就出現
Activational effects ak-tih-VAY-shun-ul	啟動作用 later hormones switch behavior on	**after puberty** 青春期以後
GnRH, LH, FSH jee-EN-AR-AYCH	性腺軸 hypothalamus-pituitary chain to gonads	**outside testosterone = levels drop** 外來睪固酮＝壓低
Oxytocin / vasopressin ok-see-TOH-sin	催產素／血管加壓素 pair bonding, parenting	**vole pair bonding** 田鼠配對

流程圖：看圖就知道循環

1. SG4｜視網膜：黑暗 vs 有光 Retina in DARK vs LIGHT （p.180 to 194, 205 to 206）



2. SG4｜視覺路徑 Visual pathway （p.180 to 194, 205 to 206）



Still recognizes FACES = ventral is normal: cross out face blindness.
Recognizes objects but NOT faces = that IS prosopagnosia.
認得出「臉」才能劃掉臉盲。認得出東西、卻認不出臉＝正是臉盲。

3. SG4｜顏色兩層 Color: trichromatic and opponent-process （p.180 to 194, 205 to 206）

Layer 1: Cones 錐細胞層 Trichromatic theory 三色說		Layer 2: Ganglion cells 神經節細胞層 Opponent-process theory 拮抗說
Red cone 紅		red vs green 紅 ↔ 綠
Green cone 綠	then →	blue vs yellow 藍 ↔ 黃
Blue cone 藍		black vs white 黑 ↔ 白

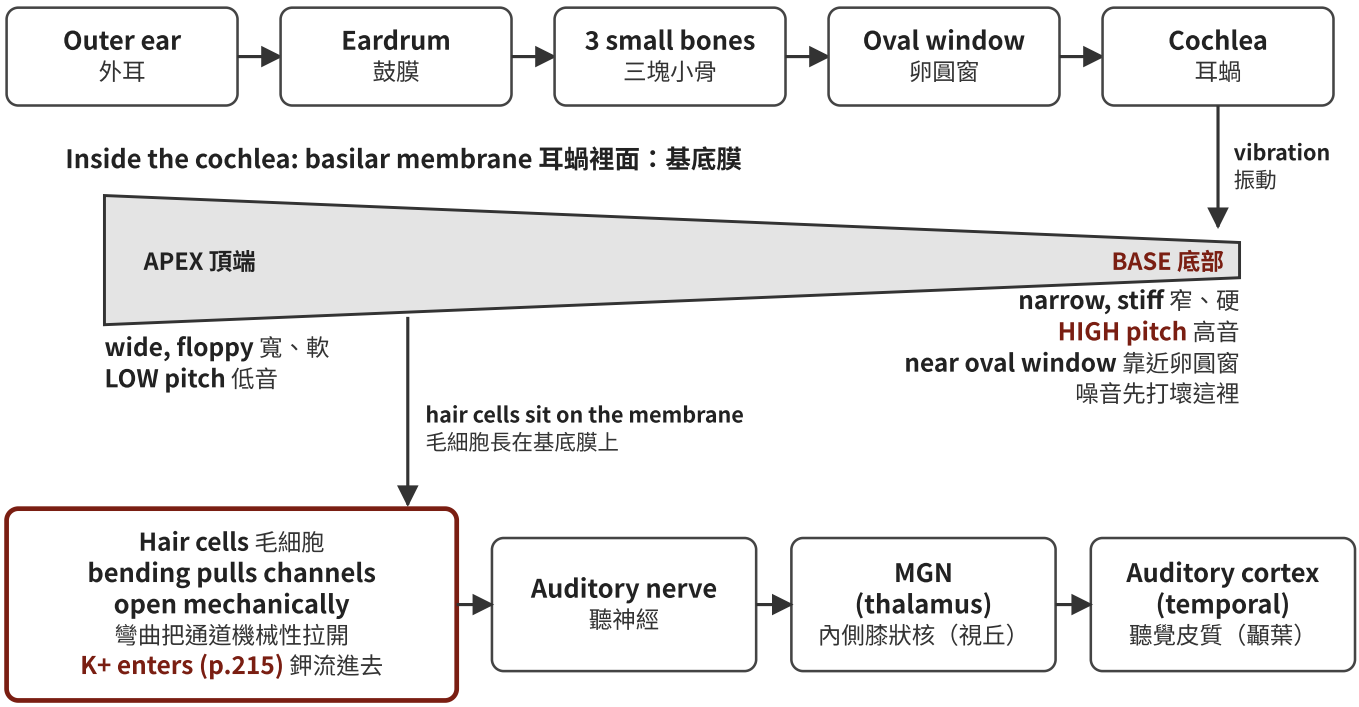
Both theories are right, at different layers. 兩個都對，只是不同層。

Afterimage 後像: the suppressed partner rebounds 一直被壓住的那一邊反彈

Stare at GREEN 盯著綠色看	→	RED partner held down 紅色那邊一直被壓住	→	Look at a white wall 移開，看白牆	→	**RED rebounds: you see red** 紅色反彈，看到紅色

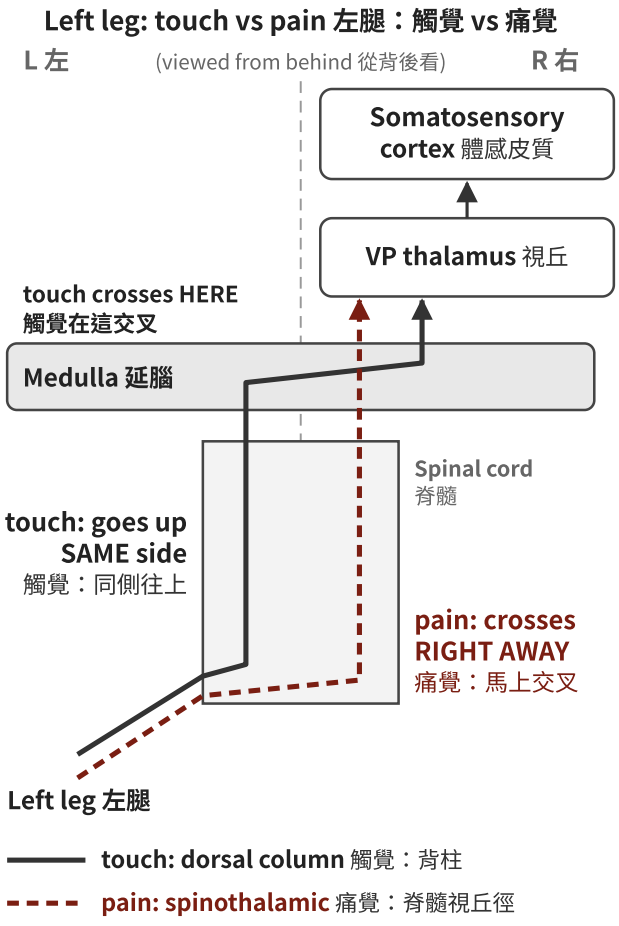
Flag example: green, black, yellow flag → you see red, white, blue on the wall.
綠黑黃國旗，看白牆變成紅白藍。

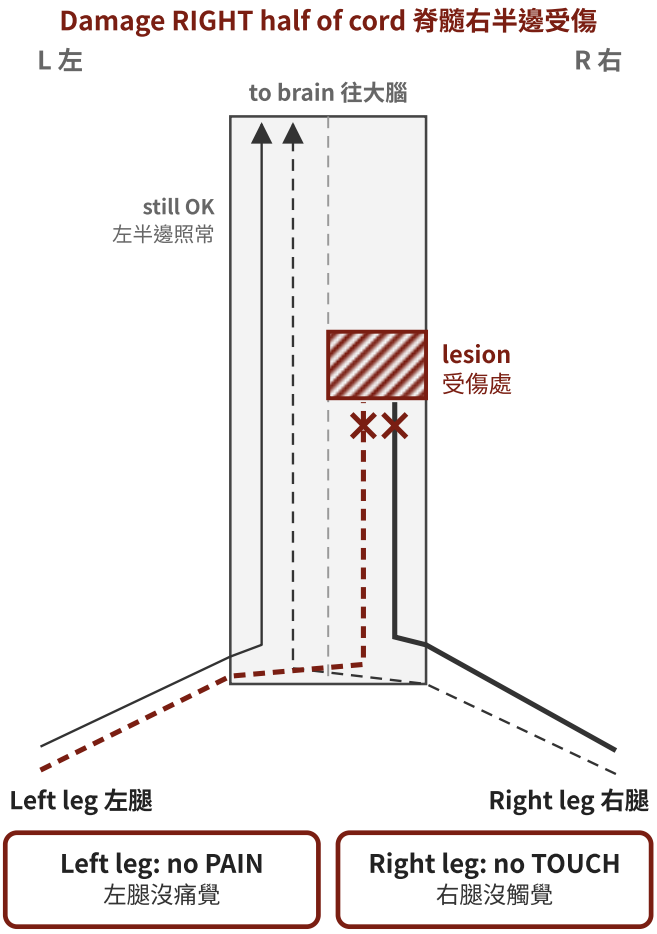
4. SG5｜聲音的路 Sound path （p.210 to 221）



First step is mechanical, not voltage. 第一步是機械性拉開，不是電壓。
Textbook p.215 says K+; professor said sodium in the 11/3 Kahoot. 選項只差 K/Na 時翻 p.215。

5. SG8｜脊髓：觸覺 vs 痛覺 Touch vs pain pathways （p.228 to 231, 235 to 236）

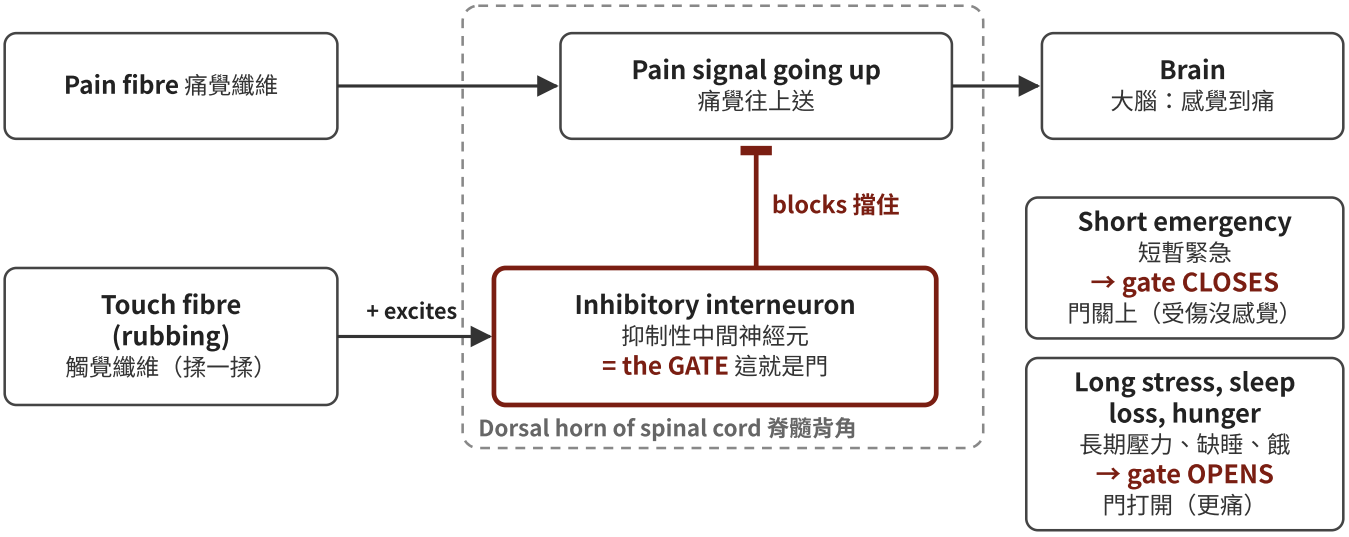




傷哪邊，哪邊沒觸覺；痛覺在另一邊

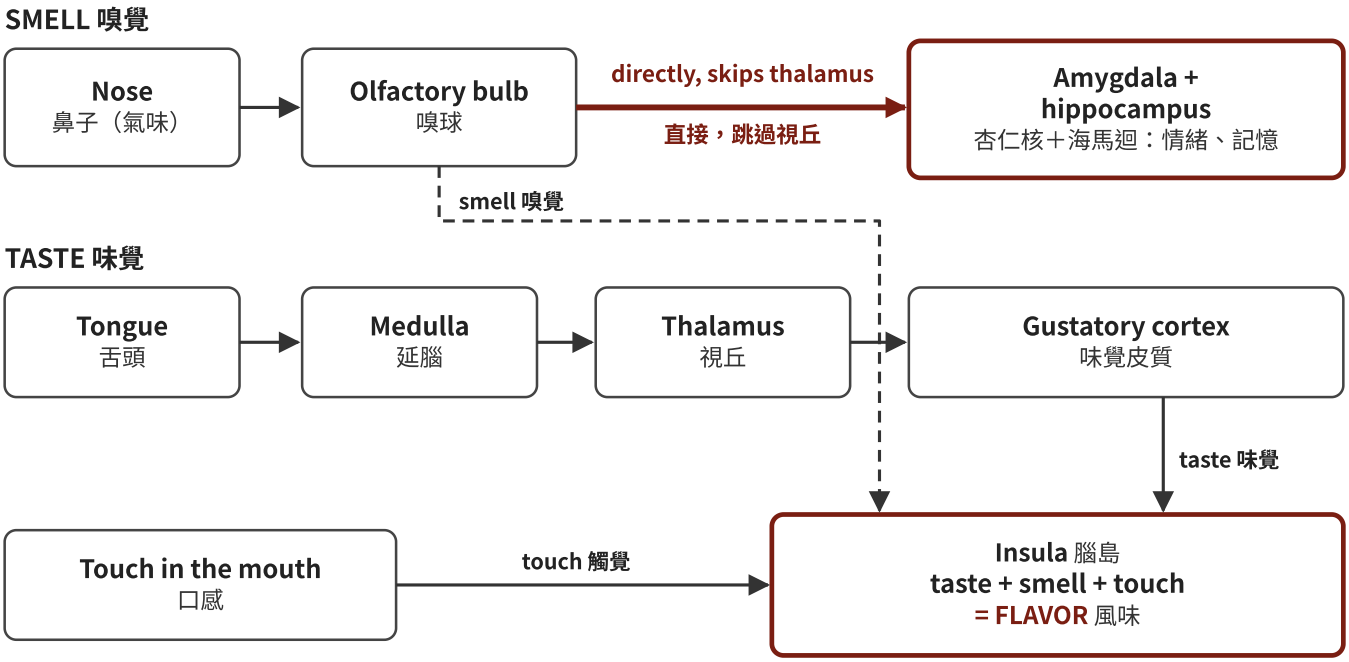
Touch is lost on the SAME side as the damage; pain is lost on the OTHER side.

6. SG10｜閘門理論 Gate theory of pain （p.235 to 237）



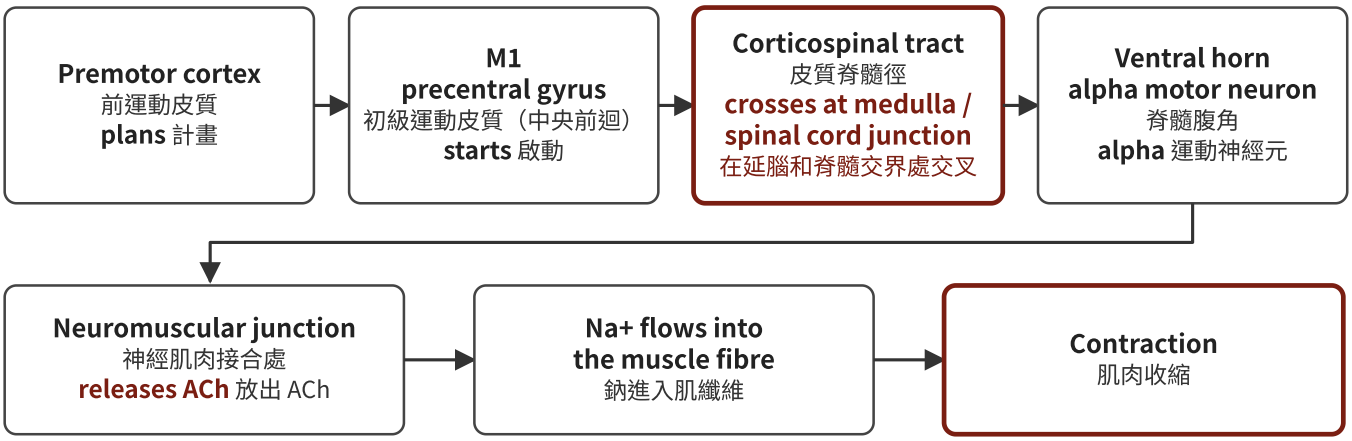
Not distraction: less pain signal really reaches the brain.
不是轉移注意力，是真的比較少痛覺訊號到大腦。

7. SG11｜嗅覺 vs 味覺 Smell vs taste （約 p.239 to 246）



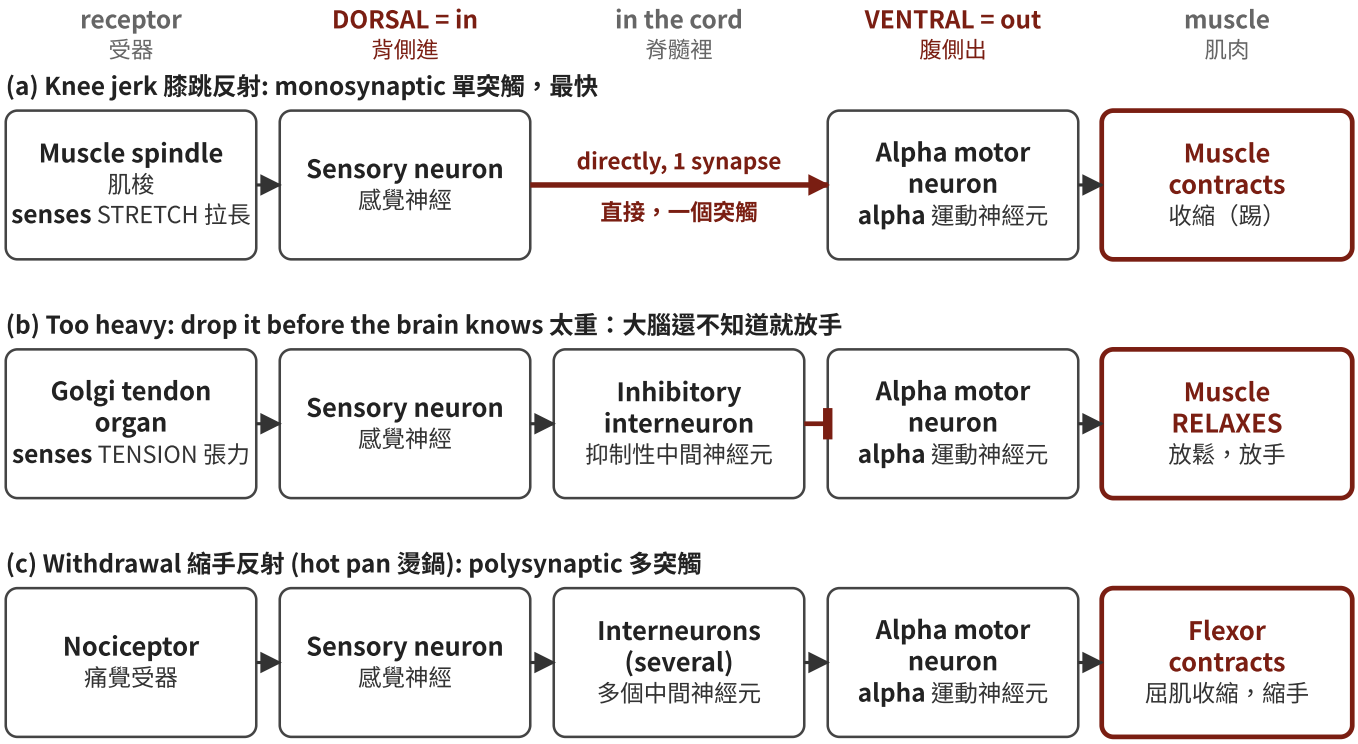
Only smell skips the thalamus; other senses stop there first. 只有嗅覺跳過視丘。
Emotional value of flavor → orbitofrontal cortex. 風味的情緒評價到眼眶額葉皮質。
Module video says insula; textbook may say orbitofrontal. 考場翻 p.239 to 246 確認。

8. SG13｜往下的運動路徑 Descending pathway （p.254 to 263）



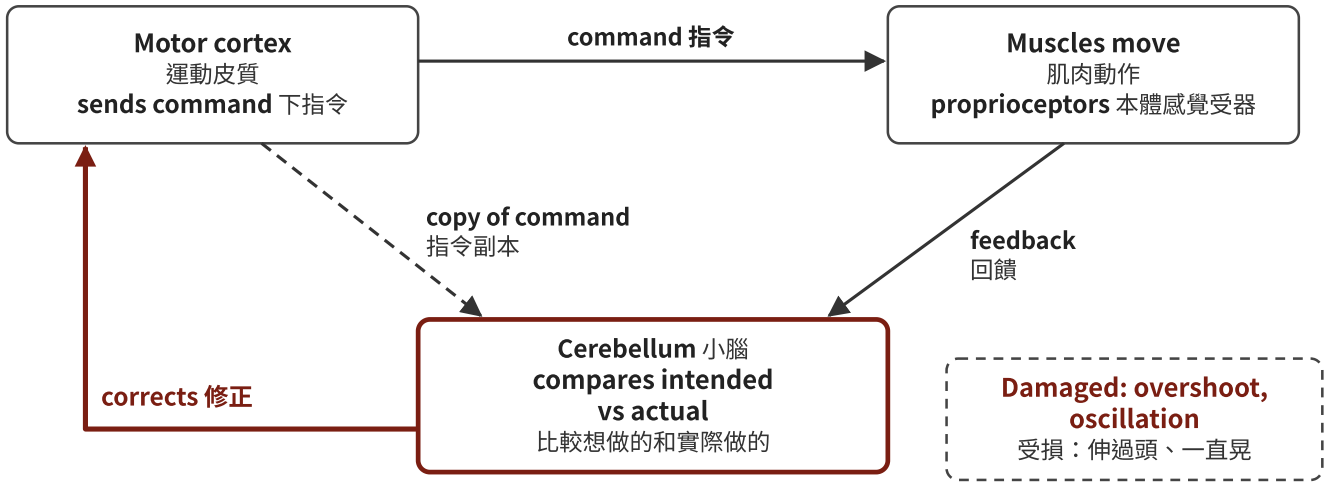
Junction uses ACh, not dopamine. 接合處是 ACh，不是多巴胺。
Asked where it CROSSES: the junction. The ventral horn is the END point. 問交叉答交界處；腹角是終點。
Left hemisphere stroke → right hand weak. 左腦中風 → 右手無力。

9. SG12 / SG13｜三種反射 Reflex arcs （p.255 to 258）



Dorsal in, ventral out. 背進腹出。
Reciprocal inhibition: biceps contracts while triceps relaxes. 互相抑制：二頭肌收縮時三頭肌放鬆。
All done in the cord: a cut cord still kicks, but he feels nothing. 脊髓斷了還會踢，但沒感覺。

10. SG15｜小腦修正迴路 Cerebellum error correction （p.263 to 264, 266）



Only the cerebellum gets BOTH the command copy and the feedback. 題目的 both 是關鍵。
Basal ganglia = gas and brakes, not quality control. 基底核是油門煞車，不是品管員。

11. SG16｜油門和煞車 Gas and brakes (basal ganglia) （p.264, 274 to 276, 279）

GAS 油門 = substantia nigra, dopamine 黑質多巴胺
BRAKES 煞車 = striatum (caudate, putamen) 紋狀體（尾核、殼核）
Steering wheel 方向盤 = motor cortex 運動皮質

Sitting still 坐著不動

GAS ▬▬▬▬▬
BRAKES ▬▬▬▬▬

balance: both at full power
平衡：兩邊都全力

not resting 不是休息
equilibrium 力量相等

Parkinson 帕金森

GAS ▬▬▭▭▭
BRAKES ▬▬▬▬▬

gas too weak, brakes normal
油門不夠，煞車正常

rest tremor 靜止會抖
hard to start 很難開始動
depression first 先憂鬱
one side first 先單側

Huntington 杭丁頓

GAS ▬▬▬▬▬
BRAKES ▬▭▭▭▭

brakes broken
煞車壞了

jerky movements 停不下來
dominant gene 顯性基因
caudate + putamen shrink
尾核、殼核萎縮

Depression first: 教授版本（10/22），黑質的多巴胺也送到快樂中樞 nucleus accumbens。
課本解剖：送到依核的主要是 VTA。

12. SG18 to 20｜體溫 Temperature and the set point （Ch9 p.282 to 311）

Set point 設定點 37°C
hypothalamus, preoptic area
下視丘，視前區

below 低於 → **Too cold** 太冷
shiver 發抖
vasoconstrict 血管收縮

above 高於 → **Too hot** 太熱
sweat 流汗
vasodilate 血管擴張

Fever 發燒

Pyrogens
致熱原（感冒）
→ **Raise the set point**
設定點被調高
→ **Normal temperature feels cold**
正常體溫被當成太冷
→ **Shivering under blankets**
發抖，蓋被子

Homeostasis = the concept; set point = the setting (37°C). 恆定是概念，設定點是具體設定。

13. SG21 / SG22 | 兩種口渴 Two thirsts and tonicity (Ch9 p.282 to 311)

OSMOTIC thirst 滲透壓口渴（吃鹹，沒失血）　**Fix: plain water** 補水就好

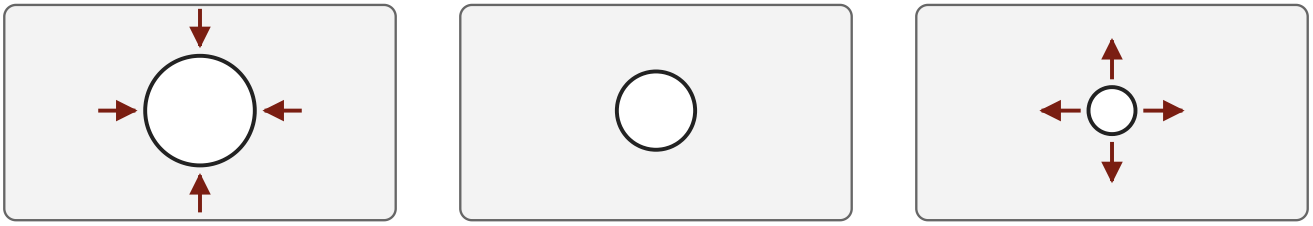
Salty food 吃鹹 → **Extracellular fluid hypertonic** 細胞外高張 → **Cells shrink** 細胞縮水 → **OVLT osmoreceptors** 滲透壓受器 → **Thirst + ADH** 口渴＋ADH

HYPOVOLEMIC thirst 低血容量口渴（失血、拉肚子）　**Needs water + salt** 要補水＋鹽

Blood loss 失血 → **Volume drops** 血量下降 → **Baroreceptors + kidneys** 壓力受器＋腎臟 → **Renin → angiotensin II** → **Subfornical organ** 穹窿下器官 → **Thirst + ADH** 口渴＋ADH

Plain water alone dilutes the blood. 只喝水會把血稀釋。　**Both thirsts feel the same.** 兩種口渴感覺一樣。

A cell placed in a solution 細胞泡在溶液裡



Hypotonic 低張
outside dilute → water IN
外面稀 → 水進去 → 脹大
例：很稀的點滴

Isotonic 等張
same as inside
一樣濃 → 不變

Hypertonic 高張
outside salty → water OUT
外面濃 → 水跑出來 → 縮水
例：3% 鹽水、吃鹹

14. SG23 / SG24 | 血糖和食慾 Glucose and feeding (Ch9 p.282 to 311)

Blood glucose 血糖（胰臟 pancreas）

High blood glucose 血糖高（吃完飯） → **Pancreas: INSULIN** 胰臟放胰島素 → **Glucose into cells, stored as glycogen** 葡萄糖進細胞、存成肝醣

Low blood glucose 血糖低（馬拉松沒吃） → **Pancreas: GLUCAGON** 胰臟放升糖素 → **Glycogen → glucose** 肝醣變回葡萄糖，大腦有糖

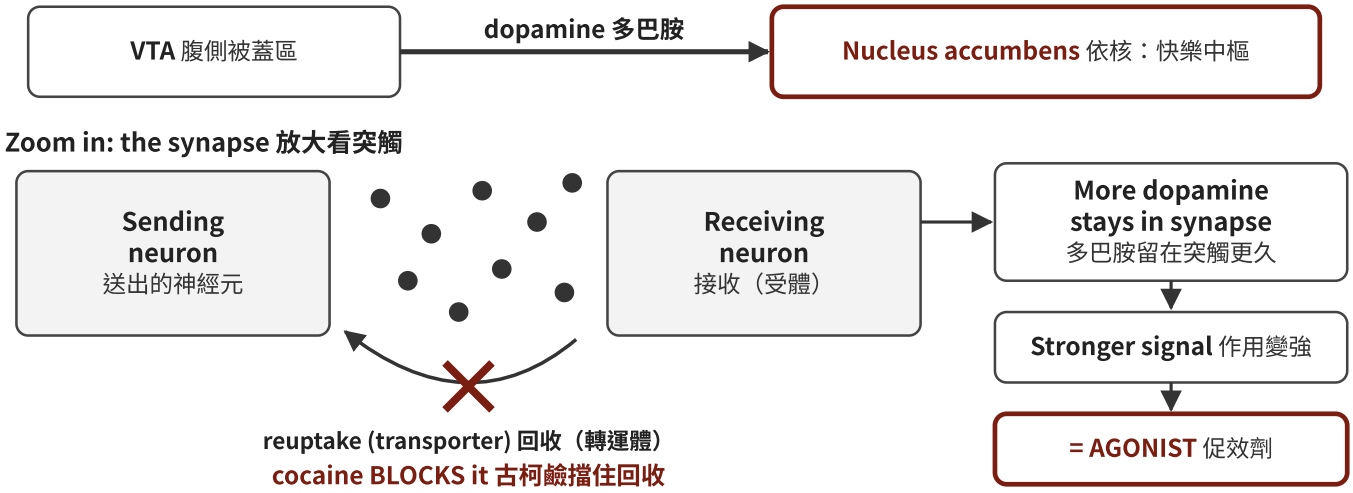
Hunger and fullness 餓和飽（下視丘 hypothalamus）

Empty stomach 空腹的胃 → **Ghrelin** 「我餓了」 → **Arcuate nucleus** 弓狀核 → **Orexin** 外側下視丘 → **EAT** 想吃

Fat cells 脂肪 → **Leptin** 「存夠了」 → **Arcuate nucleus** 弓狀核 → **STOP eating** 停，不吃

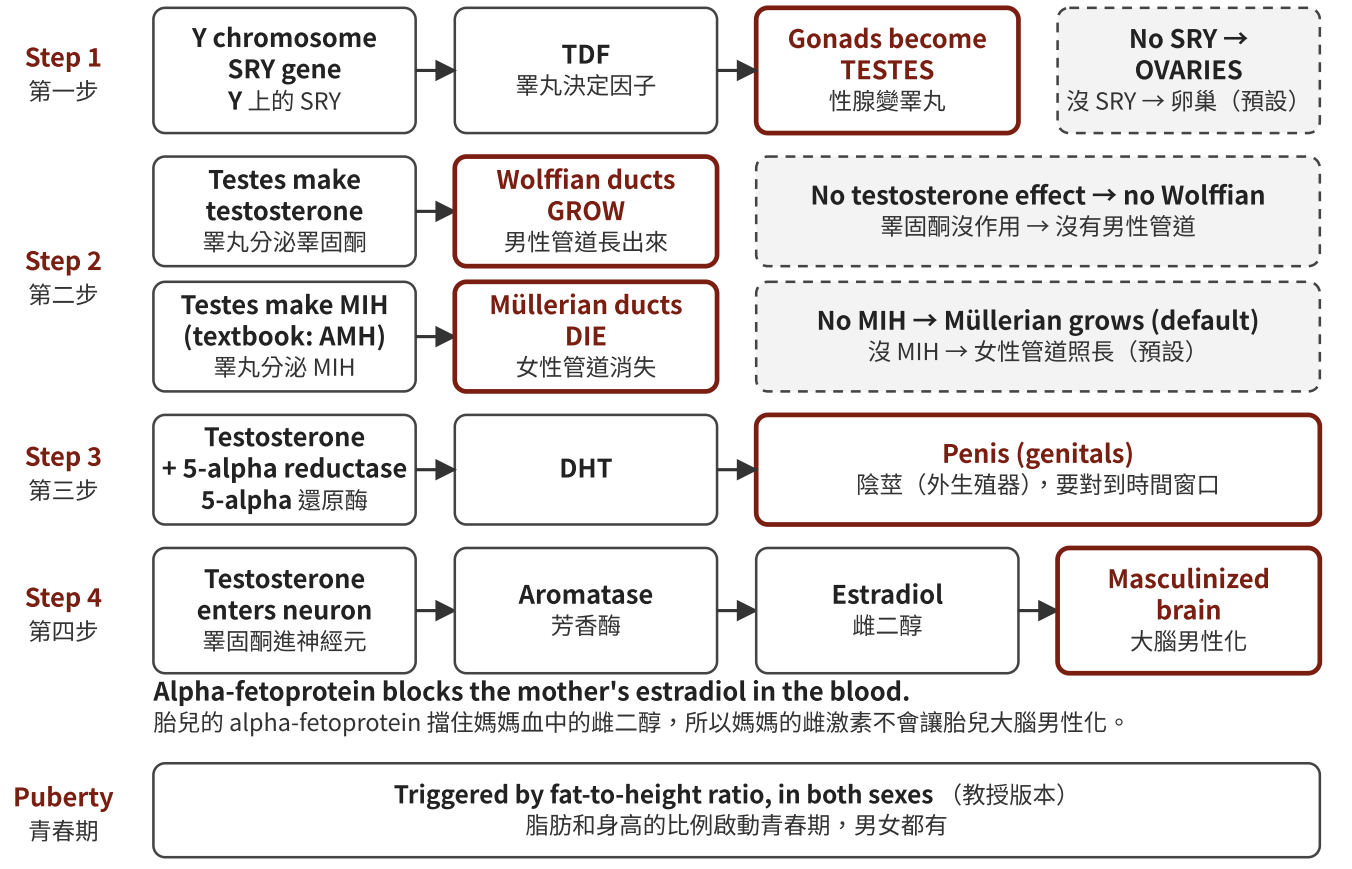
Obesity: leptin is high but the brain stops listening (leptin resistance). 胖的人 leptin 高還是餓：阻抗。
Swap trap: ghrelin = stomach, hungry; leptin = fat, full. 對調陷阱。

15. SG25｜獎賞迴路和古柯鹼 Reward and cocaine （Ch9 p.282 to 311）



It blocks reuptake, not the receptor, so the effect gets stronger. 擋的是回收，不是受體，所以作用變強。

16. SG26｜性分化四步驟 Four steps of sexual differentiation （p.314, 317 to 318, 320, 324）



Alpha-fetoprotein blocks the mother's estradiol in the blood.
胎兒的 alpha-fetoprotein 擋住媽媽血中的雌二醇，所以媽媽的雌激素不會讓胎兒大腦男性化。

Puberty 青春期 — **Triggered by fat-to-height ratio, in both sexes** （教授版本）
脂肪和身高的比例啟動青春期，男女都有

先問有沒有睪丸（MIH），再問睪固酮有沒有作用（Wolffian）

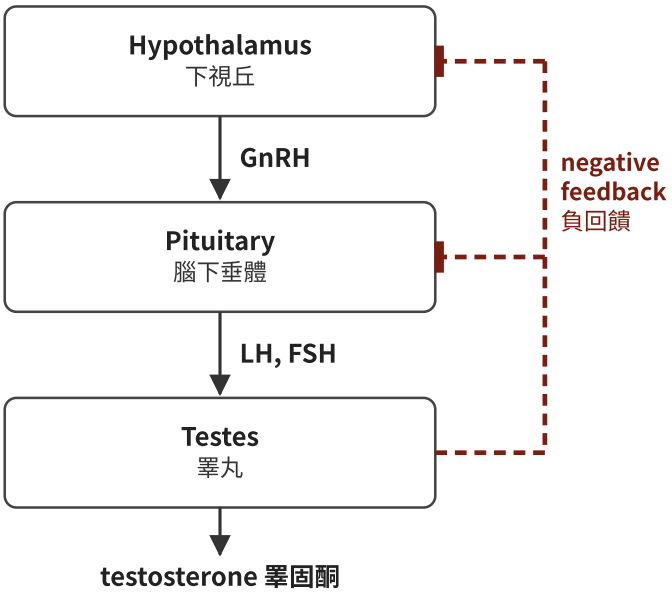
First: are there testes? (MIH kills Müllerian). Then: does testosterone work? (Wolffian grows).

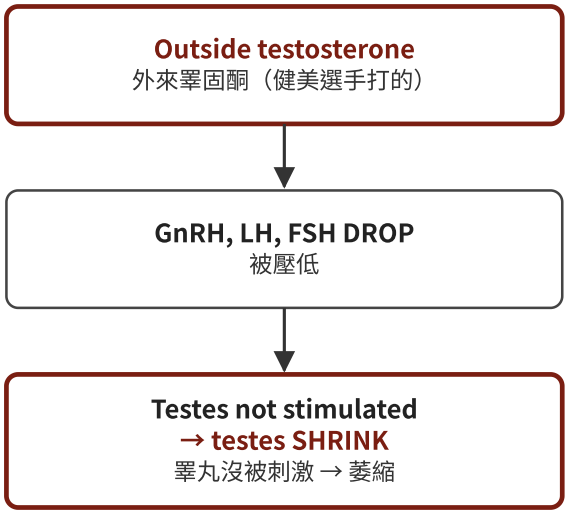
17. SG28｜發育異常對照表 Deviations （p.317 to 323）

Condition 狀況	Ducts 管道	Genitals 外生殖器	Brain 大腦	Puberty / key 青春期／關鍵
XY, testosterone has no effect (AIS) 受體沒反應	**Neither** 兩套都沒有	**Female** 女性	**Female** 女性	**Normal female** 正常女性發育 有胸部、不能生育
XY, no 5-alpha reductase 缺 5-alpha 還原酶	**Wolffian** 男性管道	**Female-looking** 外觀女性	**Masculinized** 男性化	**Masculinizes at puberty** 青春期開始男性化
XY, mutated SRY **SRY** 不作用	**Müllerian** 女性管道	**Female** 女性	**Female** 女性	**SRY = first domino** 第一張骨牌
XX, excess androgens (CAH) **XX** 大量雄性素	**Both** 兩套都有	**Masculinized** 男性化	**Masculinized** 男性化	**No testes = no MIH** 沒睪丸＝沒 MIH（課堂版本）
PMDS: MIH does not work **MIH** 沒作用	**Both** 兩套都有	**Male** 男性	**Masculinized** 男性化	**Testes yes, MIH broken** 有睪丸，MIH 壞了

Rule: testes present → MIH → Müllerian dies; testosterone works → Wolffian grows. 先問睪丸，再問睪固酮。

18. SG31｜荷爾蒙負回饋 Hormone feedback （p.337, 340, 342 to 343）





Direction: DOWN, not up. 方向：壓低，不是調高。

第六章 視覺 Vision

1. Sensation versus Perception （p.174）

After a stroke, a woman's eyes test normal. She can describe a mug's color and edges but cannot say what it is. What is impaired? [Sensation versus Perception]
中風後眼睛正常，說得出杯子的顏色和邊，但認不出是什麼。哪裡壞了？

→ **Perception is impaired; sensation is intact.**
Perception（知覺）。Sensation 正常。

Key: Can describe color and edges = the eyes transduce light, so sensation is intact; cannot name it = the brain's interpretation fails.
判斷關鍵：說得出顏色和邊＝眼睛有把光轉成訊號＝sensation 正常；認不出＝大腦詮釋壞了。

Two friends see the same dress photo on the same screen; one sees blue-black, the other white-gold. Why? [Sensation versus Perception]
同一張洋裝照片，一個看藍黑、一個看白金，為什麼？

→ **Sensation is the same, but perception differs: each brain assumes different lighting (color constancy).**
Sensation 一樣，perception 不同：大腦對光線的假設不同（color constancy）。

Key: The same light enters both people's eyes, so it is not a cone difference; the difference is how each brain interprets it.
判斷關鍵：光進眼睛一模一樣＝不是錐細胞的差別；差在大腦怎麼詮釋。

A boy with no eyes navigates by tongue clicks; scans show echoes processed in visual cortex. What does this show? [Sensation versus Perception]
沒有眼睛的男孩用彈舌回聲走路，回聲在視覺皮質處理。說明什麼？

→ **We see with the brain: the brain builds spatial perception from sound (Ben Underwood).**
We see with the brain：大腦用聲音建出空間知覺（Ben Underwood）。

Key: No eyes = no sensation of light, so the sense of space must come from the brain.
判斷關鍵：沒有眼睛＝沒有光的 sensation；空間感來自大腦。

2. Bottom-Up versus Top-Down （p.174, 196）

An author proofreads her essay three times and misses 'the the'; a stranger spots it at once. Why? [Bottom-Up versus Top-Down]
作者校稿三次沒看到 the the，陌生人一眼看到。為什麼？

→ **The author uses top-down processing: her expectation overrides what her eyes see. The stranger uses bottom-up processing.**
作者是 top-down：期待蓋過了眼睛看到的。陌生人是 bottom-up。

Key: Reading with knowledge and expectation = top-down; building up letter by letter = bottom-up, which is what catches the error.
判斷關鍵：用知識和期待去讀＝top-down；從字母一個個組起來＝bottom-up（這才抓得到錯）。

You see a word for the first time in a language you don't know, letter by letter. Which processing? [Bottom-Up versus Top-Down]
第一次看一個不認得的外文字，一個字母一個字母看。哪一種？

→ **Bottom-up processing.**
Bottom-up。

Key: A new, unexpected stimulus relies more on bottom-up processing. Both kinds run at the same time; one is just stronger.
判斷關鍵：新的、沒有預期的刺激，大腦用 bottom-up 比較多。兩種同時都在用，只是哪個比較強。

3. Photons, Light, Color, Refraction, Eye Structure （p.175–181, 198–203）

A spotlight keeps the same wavelength but sends many more photons per second. What changes? [Photons, Light, Color, Refraction, Eye Structure]
聚光燈波長不變，每秒光子變多。看起來有什麼改變？

→ **The color stays the same; it looks brighter.**
顏色不變，變亮。

Key: More photons = greater amplitude = brightness; same wavelength = same color (hue).
判斷關鍵：光子變多＝amplitude 變大＝亮度；波長沒變＝顏色沒變。

A swimmer sees blurry underwater but clearly with goggles; her eyes are healthy. Why? [Photons, Light, Color, Refraction, Eye Structure]
游泳時水裡看不清楚，戴蛙鏡就清楚，眼睛正常。為什麼？

→ **Refraction: in water, light barely bends at the cornea, so the focus misses the retina.**
Refraction：光在水裡經過角膜幾乎不彎，焦點對不到視網膜。

Key: The question is about a physical property of light = refraction; it has nothing to do with chemical changes in photoreceptors.
判斷關鍵：問的是光的物理性質＝折射；跟感光細胞的化學變化無關。

Eye drops keep the dark center opening wide in bright light. Which structure do they act on? [Photons, Light, Color, Refraction, Eye Structure]
眼藥水讓瞳孔在強光下還是很大。作用在哪個構造？

→ **The iris.**
Iris（虹膜）。

Key: The iris is the muscle (the curtain); the pupil is only an opening; the optic disc is only the back exit.
判斷關鍵：虹膜是窗簾（肌肉）；瞳孔只是洞；視神經盤只是後門。

A 55-year-old reads distant signs but holds the menu at arm's length. Why? [Photons, Light, Color, Refraction, Eye Structure]
55 歲看得清遠處，菜單要拿很遠才看得到。為什麼？

→ **The lens has stiffened and cannot thicken to focus on near objects.**
Lens（水晶體）變硬，不能變厚對焦近物。

Key: The lens is the part that adjusts; the cornea's shape is fixed.
判斷關鍵：可以調整的是水晶體；角膜形狀固定。

4. Neurons and steps in visual processing, receptive fields, visual cortex （p.180–194, 205–206）

A retina sits in total darkness. What are the photoreceptors doing? [Neurons and steps in visual processing, receptive fields, visual cortex]
視網膜在全黑中，感光細胞在做什麼？

→ **Photoreceptors are active: cGMP is high, sodium channels are open, they are depolarized, and they release glutamate constantly, inhibiting bipolar cells.**
活躍：cGMP 高、鈉通道開、去極化、一直放 glutamate，壓住雙極細胞。

Key: Professor's version: the vampire rules in the dark. "Light activates photoreceptors" is wrong.
判斷關鍵：教授版本：吸血鬼在黑暗中統治。「光活化感光細胞」是錯的。

A drug stops cGMP from being broken down. The person is in a bright room. What happens to bipolar cells? [Neurons and steps in visual processing, receptive fields, visual cortex]
藥物讓 cGMP 不被分解，人在很亮的房間。雙極細胞怎樣？

→ **Bipolar cells stay inhibited, so the person cannot see.**
一直被壓住，看不見。

Key: cGMP not broken down = photoreceptors act as if still in the dark = they keep releasing glutamate.
判斷關鍵：cGMP 不分解＝感光細胞以為還在黑暗＝繼續放 glutamate。

A faint star disappears when looked at directly but appears when looking to the side. Why? [Neurons and steps in visual processing, receptive fields, visual cortex]
暗的星星正看看不到，往旁邊看才看到。為什麼？

→ **The periphery has rods: they are very sensitive, and many rods converge on one bipolar cell.**
周邊是 rods：敏感、多個 rods 匯聚到一個雙極細胞。

Key: The center has cones, which need bright light; peripheral rods are sensitive but blurry and see no color.
判斷關鍵：中央是 cones，要亮光；周邊 rods 敏感但模糊、沒顏色。

Stare at a green-black-yellow flag, then a white wall: you see red-white-blue. Where and why? [Neurons and steps in visual processing, receptive fields, visual cortex]
盯著綠黑黃國旗再看白牆，看到紅白藍。在哪一層、為什麼？

→ **Opponent-process in ganglion cells: the side that was suppressed rebounds after you look away.**
神經節細胞的 opponent-process：一直被壓住的那一邊，移開視線後反彈。

Key: Green pairs with red, yellow with blue, black with white. Trichromatic theory (cones) is still correct; the two theories do not conflict.
判斷關鍵：綠↔紅、黃↔藍、黑↔白。三色說（錐細胞）照樣是對的，兩者不矛盾。

An on-center cell fires to a small spot on its center, but only weakly to light over its whole field. Why? [Neurons and steps in visual processing, receptive fields, visual cortex]
on-center 細胞照中心反應很強，整片照滿反應很弱。為什麼？

→ **The inhibitory surround cancels the excitatory center (lateral inhibition, horizontal cells).**
外圈抑制抵消了中心的興奮（lateral inhibition，水平細胞）。

Key: Light on the center = excitation; light on the surround = inhibition; whole field = they cancel, so the brain attends most to edges.
判斷關鍵：照中心＝興奮；照外圈＝抑制；整片＝抵消。所以大腦最注意邊緣。

After a stroke a man recognizes faces and objects, but pouring coffee looks like frozen snapshots. Damage? [Neurons and steps in visual processing, receptive fields, visual cortex]
中風後認得臉和東西，倒咖啡卻像一格一格的照片。哪裡壞了？

→ **Damage to area MT in the dorsal stream (akinetopsia).**
Dorsal stream 的 MT（akinetopsia）。

Key: Recognizes faces and objects = the ventral (what) stream is intact, so cross out prosopagnosia first. Objects OK but not faces = prosopagnosia.
判斷關鍵：題目說認得出臉和東西＝ventral（what）正常，先劃掉臉盲。只認得出東西、認不出臉＝正是臉盲。

Neuron X fires to a vertical bar at one exact spot; neuron Y fires to it anywhere, best when moving. Which is which? [Neurons and steps in visual processing, receptive fields, visual cortex]
X 只對固定位置的直線放電；Y 哪裡都行、會動最強。

→ **X is a simple cell; Y is a complex cell.**
X＝simple cell，Y＝complex cell。

Key: Fixed position = simple cell; any position and prefers motion = complex cell.
判斷關鍵：固定位置＝simple；位置不限、喜歡動＝complex。

Why can't you see the blood vessels lying in front of your retina? [Neurons and steps in visual processing, receptive fields, visual cortex]
為什麼看不到視網膜前面的血管？

→ **The stimulus is constant, so the brain filters it out and fills it in.**
刺激是恆定的，大腦把它過濾掉、補起來。

Key: Professor 11/3: constant things are not seen. Signals go one way only (photoreceptor → bipolar cell → ganglion cell), and the ganglion cell is the last layer.
判斷關鍵：教授 11/3：constant 的東西看不到。訊息單向：感光細胞 → 雙極細胞 → 神經節細胞，不能倒傳；節細胞是最後一層。

第七章 聽覺、體感、嗅味覺（p.209–246）

5. Sound waves, ear structure, auditory processing, auditory cortex （p.210–221）

A synthesizer's wave gets taller at the same rate, then repeats faster at the same height. What do listeners perceive? [Sound waves, ear structure, auditory processing, auditory cortex]
聲波先變高、頻率不變，再變快、高度不變。聽起來怎樣？

→ **First it sounds louder, then it sounds higher in pitch.**
先變大聲，再變高音。

Key: Wave height = amplitude = loudness; cycles per second = frequency = pitch.
判斷關鍵：高度＝amplitude＝響度；每秒週期數＝frequency＝音高。

After years near loud machines, an engineer can't hear high pitches but hears bass. Where is the damage? [Sound waves, ear structure, auditory processing, auditory cortex]
長年噪音，聽不到高音，低音還聽得到。哪裡壞了？

→ **The base of the basilar membrane (near the oval window).**
基底膜底部 base（靠近 oval window）。

Key: High pitch is at the base, low pitch at the apex. Professor's ribbon analogy: small waves only move the start of the ribbon.
判斷關鍵：高音在底部、低音在頂端。教授緞帶比喻：小波浪只動到開頭。

What path do the remaining sounds take to conscious hearing? [Sound waves, ear structure, auditory processing, auditory cortex]
剩下的聲音怎麼傳到意識？

→ **Auditory nerve → MGN (thalamus) → auditory cortex in the temporal lobe.**
Auditory nerve → MGN（視丘）→ 顳葉聽覺皮質。

Key: MGN = hearing; LGN is vision and VP is touch, so do not mix them up.
判斷關鍵：LGN 是視覺、VP 是觸覺，不要混。

Hair cell stereocilia bend toward the tallest one. What happens first? [Sound waves, ear structure, auditory processing, auditory cortex]
毛細胞的毛往最高那根彎，第一件發生的是什麼？

→ **Channels are pulled open mechanically, positive ions flow in, and the cell depolarizes.**
通道被機械性拉開，正離子流進去，去極化。

Key: The first step is mechanical, not voltage (professor 11/3). The textbook (p.215) says K^+ from endolymph flows in, but the 11/3 Kahoot said sodium, so if options differ only in K^+ vs Na^+, check p.215.
判斷關鍵：一開始不是電壓（教授 11/3）。離子：課本 p.215 右欄第 22 到 31 行＝內淋巴液的鉀 K^+ 流進去；教授 11/3 Kahoot 口頭說鈉。選項只差 K/Na 時翻 p.215。

6. McGurk Effect, Associative Cortices （p.219, 221）

Audio says 'ba', lips say 'ga', viewers hear 'da'; with eyes closed they hear 'ba'. What does this show? [McGurk Effect, Associative Cortices]
聲音 ba、嘴型 ga，聽到 da；閉眼聽到 ba。說明什麼？

→ **Association cortex integrates vision and hearing; vision rewrites perception.**
聯合皮質整合視覺和聽覺，視覺改寫了知覺。

Key: Eyes closed = "ba" again, so the ears did not change; the change is in the brain. You hear with your brain.
判斷關鍵：閉眼又聽到 ba＝耳朵沒變；改變在大腦。You hear with your brain。

Stopped at a red light, the car beside you rolls back and you feel yourself rolling forward. Why? [McGurk Effect, Associative Cortices]
停紅燈旁邊車往後退，覺得自己往前滑。為什麼？

→ **When the senses conflict, vision wins and overrides the vestibular sense.**
感覺衝突時視覺贏，蓋過前庭覺。

Key: "Although" the inner ear fluid did not move = the answer must explain this conflict. Neuroplasticity does not happen in seconds.
判斷關鍵：題目寫 although 內耳液體沒動＝答案要解釋這個矛盾。神經可塑性不會在幾秒內發生。

A hidden real hand and a visible rubber hand are stroked together; a hammer hits the rubber hand and she flinches. Why? [McGurk Effect, Associative Cortices]
真手藏起來和假手一起被摸，鎚子打假手她嚇到。為什麼？

→ **Vision overrides proprioception, and the brain adds the rubber hand to its body map.**
視覺蓋過本體感覺，大腦把假手收進身體地圖。

Key: Same rule as the McGurk effect, the red light illusion, and the mirror box: vision wins.
判斷關鍵：跟 McGurk、紅燈、鏡箱同一條規則。

7. Somatosensory cortex, motor cortex, receptor types （p.227–232, 235, 265）

A patient can't feel deep pressure or vibration on her fingertips but a hot mug still hurts. Why? [Somatosensory cortex, motor cortex, receptor types]
指尖感覺不到壓力和震動，摸熱杯子卻會痛。為什麼？

→ **The Pacinian corpuscles are damaged, while the separate pain receptors still work.**
Pacinian 小體壞了，獨立的痛覺受器還正常。

Key: "Whereas" is not a trap: split the option in two, and both halves must be true. Touch and pain use two different sets of receptors.
判斷關鍵：whereas 不是陷阱：切兩半，兩半都要對。觸覺和痛覺是兩組受器。

A boy has burns and fractures he never noticed; touch, reflexes, axons and myelin are normal. Cause? [Somatosensory cortex, motor cortex, receptor types]
男孩受傷都沒感覺，觸覺、反射、軸突、髓鞘都正常。原因？

→ **His nociceptors do not function.**
Nociceptors 不作用。

Key: Everything else is normal, so only the receptor at the very front is left. Pain's job is to warn.
判斷關鍵：前面全部正常＝只剩最前端的受器。痛的功能是警告。

Sudden pressure on a Pacinian corpuscle fires the fiber; steady pressure soon stops it. Why? [Somatosensory cortex, motor cortex, receptor types]
突然壓 Pacinian 小體會放電，持續壓很快就停。為什麼？

→ **Deformation pulls channels open mechanically and sodium enters; steady pressure is ignored (rapid adaptation).**
變形把通道機械性拉開、鈉進去；持續壓力它不理（快速適應）。

Key: Skin has no hair cells, and touch receptors are not opened by a neurotransmitter.
判斷關鍵：皮膚沒有毛細胞；觸覺受器不靠神經傳導物質打開。

During surgery, stimulating one strip makes the right thumb tingle; the strip in front makes it twitch. Where? [Somatosensory cortex, motor cortex, receptor types]
刺激一條皮質右拇指刺刺的，前面那條讓它抽動。在哪？

→ **Both are in the left hemisphere: tingling = postcentral gyrus (somatosensory); twitching = precentral gyrus (motor).**
都在左半腦：刺刺＝postcentral（體感），抽動＝precentral（運動）。

Key: Front = motor, back = sensory, and both control the opposite side. The two maps run in parallel, but the sensory map is set by skin input.
判斷關鍵：前面是運動，後面是感覺；都管對側。兩張地圖平行，但感覺地圖由皮膚輸入決定。

8. Vestibular sense, two-point discrimination, dermatomes, touch and pain pathways (p.228–231, 235–236)

Damage to the RIGHT half of the spinal cord at the mid-back; he can still bend his left knee. Sensory loss? [Vestibular sense, two-point discrimination, dermatomes, touch and pain pathways]
脊髓右半邊受傷，左膝還能彎。感覺怎麼失去？

→ **Loss of touch in the right leg; loss of pain in the left leg.**
右腿沒觸覺，左腿沒痛覺。

Key: Touch is lost on the injured side and pain on the opposite side. "Can bend the left knee" is a distractor, so draw it first.
判斷關鍵：傷哪邊，哪邊沒觸覺；痛覺在另一邊。「左膝能彎」是干擾。先畫圖。

Damage to the LEFT half of the thoracic cord; axons and myelin are normal. Loss below the lesion? [Vestibular sense, two-point discrimination, dermatomes, touch and pain pathways]
胸椎左半邊受傷，軸突髓鞘正常。損傷以下怎樣？

→ **Loss of touch on the left side; loss of pain on the right side.**
左邊沒觸覺，右邊沒痛覺。

Key: "Axons and myelin are normal" is a borrowed distractor and has nothing to do with the answer.
判斷關鍵：axons and myelin normal 是借來的干擾，跟答案無關。

Injury at T10: no sensation below the navel, but arms and chest are fine. Principle? [Vestibular sense, two-point discrimination, dermatomes, touch and pain pathways]
T10 受傷，肚臍以下沒感覺，手臂胸部正常。原理？

→ **Dermatomes: each area of skin maps to one level of the spinal cord, so signals from below the injury cannot go up.**
Dermatomes：每塊皮膚對應脊髓一個高度，損傷以下的上不去。

Key: Option B's weak point is "immediate": plasticity does not happen right away.
判斷關鍵：B 的破綻是 immediate：可塑性不會立刻發生。

Two caliper points feel like two on the fingertip but one on the upper arm. Why? [Vestibular sense, two-point discrimination, dermatomes, touch and pain pathways]
兩點在指尖是兩點，在上臂是一點。為什麼？

→ **Fingertips have dense receptors with small receptive fields; the upper arm has sparse receptors with large receptive fields.**
指尖受器密、感受野小；上臂受器疏、感受野大。

Key: Two points = dense receptors, small fields. Remember: where a pathway crosses depends on the type of sense (touch vs pain), not the body part.
判斷關鍵：交叉點看感覺種類（觸覺 vs 痛覺），不看身體部位。

After spinning and stopping, you still feel dizzy. Why? [Vestibular sense, two-point discrimination, dermatomes, touch and pain pathways]
轉圈停下來還是頭暈。為什麼？

→ **Fluid in the semicircular canals keeps flowing by inertia and keeps pushing the hair cells.**
半規管裡的液體有慣性，還在流動、推動毛細胞。

Key: Still dizzy after stopping = the fluid is still moving. Vestibular hair cells are mechanoreceptors too.
判斷關鍵：前庭毛細胞也是機械受器。

9. Homunculus, plasticity, phantom limbs and treatment （p.231–233）

A violinist's left-hand finger area in somatosensory cortex is enlarged. Why? [Homunculus, plasticity, phantom limbs and treatment]
小提琴家左手手指的體感皮質區變大。為什麼？

→ **Long-term use causes neuroplasticity: the area borrows neighboring neurons.**
長期使用 → 神經可塑性，借用鄰近神經元。

Key: The change is in the brain (central), not more receptors in the skin. Left hand → right hemisphere.
判斷關鍵：改變在大腦（中樞），不是皮膚受器變多。左手 → 右半腦。

Weeks after losing his right hand, stroking his right cheek feels like touch on the missing fingers. Why? [Homunculus, plasticity, phantom limbs and treatment]
右手截肢幾週後，摸右臉覺得手指被碰。為什麼？

→ **On the homunculus the face lies next to the hand; the left hemisphere's hand area gets no input and is taken over by face signals.**
小人圖上臉在手旁邊；左半腦的手區沒有輸入，被臉的訊號佔走。

Key: Right hand → left hemisphere (contralateral).
判斷關鍵：右手 → 左半腦（對側）。

An amputee's phantom hand is clenched; seeing the intact hand open in a mirror box relieves it. Why? [Homunculus, plasticity, phantom limbs and treatment]
幻肢一直握拳，鏡箱看到手張開就緩解。為什麼？

→ **Vision supplies the missing feedback, so the motor cortex stops sending the clench command.**
視覺補上缺少的回饋，運動皮質停止一直下令握拳。

Key: No feedback = a positive feedback loop that clenches tighter and tighter. The missing limb has no tendon, so it cannot activate the Golgi tendon organ.
判斷關鍵：沒有回饋＝正回饋迴圈越握越緊。斷肢沒有肌腱，不會啟動 Golgi tendon organ。

10. Gate theory of pain （p.235–237）

After bumping an elbow, rubbing it lessens the pain. How? [Gate theory of pain]
撞到手肘揉一揉比較不痛。怎麼做到的？

→ **Touch signals activate inhibitory interneurons in the spinal cord, which block ascending pain (the gate closes).**
觸覺訊號啟動脊髓的抑制性中間神經元，擋住往上的痛覺（門關上）。

Key: It is not distraction; less pain signal actually reaches the brain. Rubbing does not destroy pain nerve endings.
判斷關鍵：不是轉移注意力，是真的比較少痛覺到大腦。揉一揉不會破壞痛覺神經末梢。

A player doesn't notice a deep cut during a game; a week of finals stress makes a paper cut hurt more. Why? [Gate theory of pain]
比賽中受傷沒感覺，期末壓力一週後紙割傷特別痛。為什麼？

→ **A short-term emergency closes the gate; long-term stress opens the gate.**
短暫緊急把門關上；長期壓力把門打開。

Key: Do not reverse the direction: emergency = gate closed, chronic stress = gate open.
判斷關鍵：方向不要對調。

Pain is worse on days of sleep loss and stress; nerves and healing are normal. Mechanism? [Gate theory of pain]
缺睡、壓力大的日子更痛，神經和傷口都正常。機制？

→ **Descending inhibitory pathways weaken, so more pain signals reach the brain.**
往下的抑制路徑變弱，更多痛覺訊號上到大腦。

Key: The question says peripheral nerves are normal = cross out peripheral options first. Receptors do not change by themselves.
判斷關鍵：題目說周邊神經正常＝先劃掉周邊選項。受器不會自己變身。

11. Smell and taste pathways （約 p.239–246）

Smelling lavender brings a wave of childhood emotion before she can name the scent. Why so fast? [Smell and taste pathways]
聞到薰衣草，還沒認出就湧上童年情緒。為什麼這麼快？

→ **Smell bypasses the thalamus and goes directly to the amygdala and hippocampus.**
嗅覺跳過視丘，直通杏仁核和海馬迴。

Key: Senses in the head do not use the spinal cord. Every other sense relays through the thalamus first.
判斷關鍵：頭上的感覺都不走脊髓。其他感覺都要先到視丘轉接。

With a pinched nose, she tastes sweetness but can't tell apple from pear. Where is flavor built? [Smell and taste pathways]
捏著鼻子只嚐到甜，分不出蘋果和梨。風味在哪合成？

→ **The insula, where taste, smell, and touch meet; emotional value goes to the orbitofrontal cortex. (Module video version; the textbook may say orbitofrontal, so check p.239 to 246.)**
Insula（腦島），味覺、嗅覺、觸覺會合；情緒評價到 orbitofrontal cortex。（Module 影片版本；課本可能寫 orbitofrontal，考場翻 p.239 到 246 確認）

Key: The tongue tastes only basic tastes; flavor = taste + smell + mouthfeel.
判斷關鍵：舌頭只嚐基本味；風味＝味覺＋嗅覺＋口感。

第八章 動作系統（p.248–279）

12. Ascending pathway from proprioceptors （p.255–258）

A patient moves his legs on command but stumbles in the dark and can't tell where his feet are. What is lost? [Ascending pathway from proprioceptors]
叫他動腳動得了，暗處卻不知道腳在哪。失去什麼？

→ **Ascending proprioception is lost (muscle spindles and Golgi tendon organs to the cerebellum and cortex).**
往上的本體感覺（肌梭、Golgi tendon organ 到小腦和皮質）。

Key: Can move on command = the descending pathway is intact; cannot tell where the feet are = the ascending signal is lost. (p.255 to 258)
判斷關鍵：能照指令動＝往下的路正常。（p.255–258）

13. Descending pathway, motor neurons, acetylcholine, reflexes （p.254–263）

The motor cortex and alpha motor neurons fire normally, yet muscles barely contract. Which step fails? [Descending pathway, motor neurons, acetylcholine, reflexes]
皮質和 alpha 運動神經元都正常，肌肉卻幾乎不收縮。哪一步壞？

→ **ACh cannot bind to the muscle's receptors, so the sodium channels do not open.**
ACh 接不上肌肉的受器，鈉通道打不開。

Key: The neuromuscular junction uses ACh, not dopamine. (p.254)
判斷關鍵：神經肌肉接合處是 ACh，不是 dopamine。（p.254）

A left-hemisphere stroke weakens the right hand. Where did those axons cross? [Descending pathway, motor neurons, acetylcholine, reflexes]
左半腦中風右手無力，軸突在哪交叉？

→ **At the junction of the medulla and spinal cord (pyramidal decussation).**
延腦和脊髓交界處（pyramidal decussation）。

Key: Asked where they cross = the junction; the ventral horn is the end point; the midbrain is the rubrospinal tract. (p.263)
判斷關鍵：問 cross 答交界處；腹角是終點；midbrain 是 rubrospinal。（p.263）

A climber suddenly lets go of a ledge before deciding to, when forearm strain is extreme. Mechanism? [Descending pathway, motor neurons, acetylcholine, reflexes]
攀岩者手臂張力太大，還沒決定就放手。機制？

→ **Golgi tendon organ → inhibitory interneuron → inhibits alpha motor neurons.**
Golgi tendon organ → 抑制性中間神經元 → 抑制 alpha 運動神經元。

Key: Tension = Golgi tendon organ; stretch = muscle spindle, which makes the muscle contract. (p.258)
判斷關鍵：張力＝tendon organ；拉長＝spindle（讓肌肉收縮）。（p.258）

Complete mid-back cord injury: the knee kicks to a tap but he feels nothing. Why? [Descending pathway, motor neurons, acetylcholine, reflexes]
胸椎完全斷，敲膝蓋會踢但沒感覺。為什麼？

→ **The knee jerk is monosynaptic and happens entirely in the spinal cord; sensation cannot travel up to the brain.**
膝跳是單突觸，整個在脊髓完成；感覺上不去大腦。

Key: "Is the reflex there?" and "Can he feel it?" have opposite answers. (p.256)
判斷關鍵：反射有沒有、感覺得到嗎，答案相反。（p.256）

Touching a hot pan jerks the arm back; a tendon tap kicks the leg. How do the circuits differ? [Descending pathway, motor neurons, acetylcholine, reflexes]
碰燙鍋縮手 vs 敲膝蓋踢腳，迴路差在哪？

→ **Withdrawal = nociceptors, polysynaptic; knee jerk = muscle spindle, monosynaptic.**
縮手＝痛覺受器、多突觸；膝跳＝肌梭、單突觸。

Key: Hot pan = pain receptor plus interneurons; tendon tap = stretch receptor, one synapse. Option D's trap swaps the receptors.
判斷關鍵：D 的陷阱是感受器對調。

The biceps contracts while the triceps relaxes. What spinal mechanism? [Descending pathway, motor neurons, acetylcholine, reflexes]
二頭肌收縮時三頭肌放鬆，什麼機制？

→ **Reciprocal inhibition.**
Reciprocal inhibition。

Key: Muscles can only pull, not push; if both sides pulled at once, the joint would lock. (p.259)
判斷關鍵：肌肉只會拉不會推，兩邊同時拉就卡住。(p.259)

Fine eye muscles vs the calf: how do motor units differ? [Descending pathway, motor neurons, acetylcholine, reflexes]
眼睛肌肉 vs 小腿肌肉，運動單位差在哪？

→ **Fine control = small motor units (one neuron controls few muscle fibers); force = large motor units.**
精細＝小單位（一個神經元管很少肌纖維）；出力＝大單位。

Key: Precise = small units, powerful = large units. Exact numbers are not tested. (p.254)
判斷關鍵：不考數字。(p.254)

14. Somatotopic map, cortically controlled FES, treating paralysis （p.265–266）

Stimulating a small strip moves fingers and lips; an equal strip moves the whole leg. Principle? [Somatotopic map, cortically controlled FES, treating paralysis]
一小塊皮質動手指和嘴唇，同樣大小動整條腿。原理？

→ **Somatotopic map: body parts with finer control get more cortex.**
Somatotopic map：越精細的部位分到越多皮質。

Key: Cortex area follows precision of control, not body size.
判斷關鍵：面積看精細度，不看身體大小。

A man with a neck cord injury picks up a cup using a motor-cortex implant that stimulates forearm muscles. What is true? [Somatotopic map, cortically controlled FES, treating paralysis]
頸椎受傷的人用運動皮質晶片刺激前臂拿杯子。哪個對？

→ **FES reads out the signal and bypasses the damaged cord, restoring some movement; it does not repair the injury.**
FES：讀出訊號、繞過受損脊髓，恢復部分動作，不修復損傷。

Key: FES reads signals OUT; DBS sends stimulation IN to the brain.
判斷關鍵：FES 讀出 OUT；DBS 送進 IN。

15. Cerebellum in error correction （p.263–264, 266）

Reaching for a cup, his hand oscillates and overshoots repeatedly. Which structure gets both command copies and feedback? [Cerebellum in error correction]
伸手拿杯子一直晃、伸過頭。哪個構造同時收指令副本和回饋？

→ **The cerebellum.**
Cerebellum。

Key: "Both" is the key word, since the somatosensory cortex receives only feedback. Overshoot = think cerebellum first. (p.263 to 264)
判斷關鍵：題目的 both 是關鍵；體感皮質只收回饋。看到 overshoot 先想小腦。（p.263–264）

With prism glasses, a healthy person adapts after a few throws; a patient keeps missing. Damage? [Cerebellum in error correction]
戴稜鏡丟飛鏢，正常人幾次就修正，病人一直偏。哪裡壞？

→ **The cerebellum (it corrects errors trial by trial).**
Cerebellum（一次一次修正錯誤）。

Key: Adapts after a few throws = error correction = cerebellum. The basal ganglia are the gas and brakes, not the quality checker.
判斷關鍵：基底核是油門煞車，不是品管員。

16. Motor disorders (Huntington and Parkinson) （p.264, 274–276, 279）

A 62-year-old's hand trembles at rest and he freezes in doorways. Gas and brakes? [Motor disorders (Huntington and Parkinson)]
62 歲靜止時手抖、走到門口卡住。油門煞車怎樣？

→ **Not enough gas (low dopamine from the substantia nigra), brakes normal: Parkinson's disease.**
油門不夠（黑質多巴胺少），煞車正常 → Parkinson。

Key: Tremor at rest = brakes overpower the gas; hard to start moving = brakes released but still no gas.
判斷關鍵：坐著抖＝煞車蓋過油門；要動更難＝放開煞車還是沒油。

A woman in her forties with jerky movements she can't stop; her caudate and putamen have shrunk. Cause? [Motor disorders (Huntington and Parkinson)]
四十幾歲抽動停不下來，尾核和殼核萎縮。原因？

→ **Degeneration of the striatum (the brakes): Huntington's disease, a dominant gene.**
Striatum（煞車）退化 → Huntington，顯性基因。

Key: Caudate + putamen = striatum = brakes.
判斷關鍵：caudate＋putamen＝striatum＝煞車。

A man felt flat and joyless for two years before a resting tremor started. Why did mood change first? [Motor disorders (Huntington and Parkinson)]
先兩年憂鬱，後來才手抖。為什麼情緒先出現？

→ **Professor's version (10/22): substantia nigra dopamine also goes to the pleasure center, the nucleus accumbens, so the pleasure gas runs out first. (Textbook anatomy: the main input to the nucleus accumbens is from the VTA.)**
教授版本（10/22）：黑質的多巴胺也送到快樂中樞 nucleus accumbens，快樂的油門先沒了。（課本解剖：送到依核的主要是 VTA）

Key: Mood before tremor = dopamine to the reward center drops first. It is not the striatum suppressing the prefrontal cortex.
判斷關鍵：不是紋狀體去壓前額葉。

A student sits completely still. What are the basal ganglia doing? [Motor disorders (Huntington and Parkinson)]
學生坐著不動，基底核在做什麼？

→ **The gas and brakes are both working at full force, equally balanced (equilibrium).**
油門和煞車都在全力運作，力量相等（equilibrium）。

Key: Sitting still is not rest.
判斷關鍵：坐著不動不是休息。

Cortex and muscle are normal, but motor axon signals are slow and out of step. Problem? [Motor disorders (Huntington and Parkinson)]
皮質和肌肉都正常，軸突訊號又慢又散。問題？

→ **Demyelination (for example, MS).**
Demyelination（例如 MS）。

Key: Slow, out-of-step signals = myelin. Cross out whatever the question calls normal first; hierarchy: muscle, myelin, spinal cord, basal ganglia, cortex.
判斷關鍵：題目說正常的先劃掉。階層：肌肉 → 髓鞘 → 脊髓 → 基底核 → 皮質。

Why does Parkinson often start on one side of the body? [Motor disorders (Huntington and Parkinson)]
帕金森為什麼常從一側開始？

→ **Each side of the brain has its own basal ganglia, and neurons on one side may die first.**
兩側各有基底核，一側的神經元可能先死。

Key: "The pathway crosses at the medulla" is the right fact used in the wrong place.
判斷關鍵：交叉點在延腦這個說法用錯地方。

17. Mirror neurons （p.266）

A musician sits motionless watching a pianist; his motor neurons fire in finger patterns. Which neurons? [Mirror neurons]
音樂家坐著不動看鋼琴家，運動區神經元照手指模式放電。是什麼？

→ **Mirror neurons: they fire both when you act and when you watch someone else act.**
Mirror neurons：自己做和看別人做都放電。

Key: Motionless = his hands do not move, so it is not alpha motor neurons. The link to autism is only speculation, not proven.
判斷關鍵：motionless＝手沒動，所以不是 alpha 運動神經元。和自閉症的關聯只是推測，不是證實。

第九章 恆定、動機、獎賞（p.282–311）

18. Drive reduction vs reward, homeostasis, set point （Ch9 p.282–311）

Sam eats plain crackers until hunger goes away; later, full, he orders chocolate cake. Difference? [Drive reduction vs reward, homeostasis, set point]
餓了吃餅乾到不餓，飽了還點蛋糕。差在哪？

→ **Crackers = drive reduction (push); cake = reward (pull).**
餅乾＝drive reduction（推）；蛋糕＝reward（拉）。

Key: Eating until hunger is gone = reducing a need (push); eating when full = an incentive (pull). Both are motivation.
判斷關鍵：兩者都是動機。

Which is the setting, and which is the concept: 37°C or keeping temperature stable? [Drive reduction vs reward, homeostasis, set point]
37 度和維持體溫穩定，哪個是設定值、哪個是概念？

→ **37°C = set point; keeping temperature stable = homeostasis.**
37°C＝set point；維持穩定＝homeostasis。

Key: Professor 11/3: homeostasis is the concept; the set point is the setting.
判斷關鍵：教授 11/3：homeostasis 是概念，set point 是設定。

19. Body temperature, endotherm vs ectotherm, hyper- vs hypothermia （Ch9 p.282–311）

A lizard basks before moving fast; a squirrel holds its temperature; a runner's core temperature climbs. Labels? [Body temperature, endotherm vs ectotherm, hyper- vs hypothermia]
蜥蜴曬太陽、松鼠體溫穩定、跑者體溫飆高。各是什麼？

→ **Lizard = ectotherm; squirrel = endotherm; runner = hyperthermia.**
蜥蜴 ectotherm、松鼠 endotherm、跑者 hyperthermia。

Key: Basks in the sun = depends on the environment; holds its own temperature = makes its own heat. Hyper = too much, hypo = too little.
判斷關鍵：hyper 太多、hypo 太少。

20. Temperature-sensitive neurons of the hypothalamus （Ch9 p.282–311）

Early flu: normal temperature, yet she shivers under blankets feeling freezing. Why? [Temperature-sensitive neurons of the hypothalamus]
感冒初期體溫正常，卻蓋三條被子發抖。為什麼？

→ **Pyrogens raise the hypothalamus's set point, so a normal temperature is read as too cold.**
Pyrogens 把下視丘設定點調高，正常體溫被當成太冷。

Key: Shivering is a defense against cold; warm-sensitive neurons handle cooling.
判斷關鍵：發抖是冷的防禦；warm-sensitive 神經元負責散熱。

21. Fluids, osmosis, isotonic, hypotonic, hypertonic （Ch9 p.282–311）

A patient gets an IV of a very dilute solution. What happens to his cells? [Fluids, osmosis, isotonic, hypotonic, hypertonic]
點滴是很稀的溶液，細胞怎樣？

→ **Hypotonic: water moves into the cells, and they swell.**
Hypotonic：水進細胞，細胞脹大。

Key: Dilute outside = hypotonic = water moves in. Only hypertonic makes cells shrink.
判斷關鍵：外面稀＝低張＝水往裡跑。高張才縮水。

Cells are placed in 3% saline. What happens? [Fluids, osmosis, isotonic, hypotonic, hypertonic]
細胞放進 3% 鹽水，會怎樣？

→ **Hypertonic: water moves out of the cells, and they shrink.**
Hypertonic：水跑出來，細胞縮水。

Key: Turn the scenario into the term: 3% saline → hypertonic → shrink.
判斷關鍵：情境轉術語：3% saline → hypertonic → shrink。

22. Hypovolemic vs osmotic thirst, osmoreceptors （Ch9 p.282–311）

Minutes after salty pretzels, with no sweating or blood loss, you're very thirsty. Sequence? [Hypovolemic vs osmotic thirst, osmoreceptors]
吃完鹹餅乾很渴，沒流汗沒失血。順序？

→ **Hypertonic extracellular fluid → cells shrink → OVLT osmoreceptors → thirst + ADH.**
細胞外高張 → 細胞縮水 → OVLT 滲透壓受器 → 口渴＋ADH。

Key: Salty food with no blood loss = osmotic thirst, not hypovolemic.
判斷關鍵：沒失血＝不是低血容量。

A soldier loses a lot of blood and is thirsty; he gets only plain water. Which is correct? [Hypovolemic vs osmotic thirst, osmoreceptors]
士兵失血很渴，只給白開水。哪個對？

→ **Hypovolemic thirst (baroreceptors, kidneys, angiotensin II → subfornical organ); he needs water plus salt.**
Hypovolemic thirst（baroreceptors、腎臟、angiotensin II → subfornical organ）；要補水＋鹽。

Key: Blood loss removes water and salt together, so plain water dilutes the blood. The two kinds of thirst feel the same.
判斷關鍵：失血是水和鹽一起少；只補水會把血稀釋。兩種口渴感覺一樣。

23. Glucose, glucoreceptors, insulin and glucagon （Ch9 p.282–311）

Hours into a marathon without food, her blood glucose falls but her brain still gets glucose. Which response? [Glucose, glucoreceptors, insulin and glucagon]
馬拉松沒吃東西，血糖下降，大腦還有葡萄糖。哪個反應？

→ **The pancreas releases glucagon, which turns glycogen back into glucose.**
Pancreas 放 glucagon，肝醣變回葡萄糖。

Key: Low blood sugar = glucagon; insulin stores glucose (swap trap).
判斷關鍵：血糖低＝升糖素；胰島素是存起來（對調陷阱）。

Right after a big meal, which hormone moves glucose into cells? [Glucose, glucoreceptors, insulin and glucagon]
大吃一頓後，哪個荷爾蒙把葡萄糖送進細胞？

→ **Insulin.**
Insulin。

Key: High blood sugar = insulin.
判斷關鍵：血糖高＝胰島素。

24. Fat, leptin, ghrelin, orexins, feeding （Ch9 p.282–311）

Before lunch, her stomach is empty and a bakery smell makes her hungry. Mechanism? [Fat, leptin, ghrelin, orexins, feeding]
午餐前空腹，聞到麵包香很餓。機制？

→ **Ghrelin (stomach) → arcuate nucleus → orexin neurons → hunger.**
Ghrelin（胃）→ 弓狀核 → orexin 神經元 → 想吃。

Key: Empty stomach = ghrelin; leptin is fat saying "enough" (swap trap).
判斷關鍵：leptin 是脂肪說夠了（對調陷阱）。

Leptin-less mice overeat and leptin shots help them, yet many obese people have high leptin and stay hungry. Why? [Fat, leptin, ghrelin, orexins, feeding]
缺 leptin 的老鼠打了會少吃，但胖的人 leptin 高還是餓。為什麼？

→ **Leptin from fat tells the hypothalamus that stores are full; many obese people develop leptin resistance.**
Leptin 從脂肪告訴下視丘存夠了；很多胖的人產生阻抗。

Key: High leptin but still hungry = resistance. The hunger signal from the stomach is ghrelin.
判斷關鍵：飢餓訊號從胃來的是 ghrelin。

25. Reward circuits, cocaine （Ch9 p.282–311）

A drug blocks dopamine transporters in the nucleus accumbens. Effect? [Reward circuits, cocaine]
藥物擋住依核的多巴胺轉運體，效果？

→ **Dopamine stays in the synapse longer, strengthening VTA → NAc; the drug is an agonist.**
多巴胺留在突觸更久，強化 VTA → NAc，是 agonist。

Key: It blocks reuptake, not the receptor, so the effect gets stronger.
判斷關鍵：擋的是回收，不是受體，所以作用變強。

A rat always got a drug in a green box; the same dose in a red box killed it. Why? [Reward circuits, cocaine]
老鼠一直在綠盒拿藥，紅盒同劑量死了。為什麼？

→ **The brain prepares in advance based on context; without the familiar cue there is no preparation, so the same dose becomes an overdose.**
大腦根據情境預先準備；沒有熟悉線索就沒準備，同劑量變過量。

Key: Same dose, new box = no learned preparation. Professor 10/27 class: addiction is a disease, not a choice.
判斷關鍵：教授 10/27 課堂。成癮是疾病，不是選擇。

第十章 性別與性分化（p.314 起）

26. Sex vs gender, 4 steps of sexual differentiation （p.314, 317–318, 320, 324）

An XY person's SRY gene doesn't work. Ducts, genitals, brain? [Sex vs gender, 4 steps of sexual differentiation]
XY 但 SRY 不作用。管道、外生殖器、大腦？

→ **Müllerian ducts, female genitals, female brain: fully female.**
Müllerian、女性、女性：完全女性。

Key: SRY is the first domino; the main job of the Y chromosome is SRY.
判斷關鍵：SRY 是第一張骨牌；Y 的主要工作就是 SRY。

Why doesn't a pregnant mother's high estradiol masculinize her XX fetus's brain? [Sex vs gender, 4 steps of sexual differentiation]
媽媽雌二醇很高，為什麼不會讓 XX 胎兒大腦男性化？

→ **Alpha-fetoprotein blocks estradiol in the blood; testosterone becomes estradiol through aromatase only after it enters neurons.**
Alpha-fetoprotein 擋住血中的雌二醇；睪固酮進神經元後才被 aromatase 轉成雌二醇。

Key: The brain is masculinized by estradiol (aromatase); the external genitals by DHT.
判斷關鍵：大腦靠 estradiol（aromatase）；外生殖器靠 DHT。

An XY boy has plenty of testosterone but no aromatase in his neurons. Brain? [Sex vs gender, 4 steps of sexual differentiation]
XY 男孩睪固酮很多，但神經元沒有 aromatase。大腦？

→ **A fully female brain.**
完全女性。

Key: Professor 11/3: testosterone is only a pass; it must become estrogen inside the cell to masculinize the brain.
判斷關鍵：教授 11/3：睪固酮只是通行證，要在細胞裡變成雌激素才男性化。

A character has a male body, identifies as a woman, and is attracted to women. What point? [Sex vs gender, 4 steps of sexual differentiation]
男性身體、認同女性、喜歡女性。說明什麼？

→ **Biological sex, gender identity, and sexual orientation are three separate things.**
生理性別、性別認同、性傾向是三件分開的事。

Key: "Always" and "determines" are absolute words, so those options are wrong.
判斷關鍵：always、determines 是絕對字。

What triggers puberty in both sexes? [Sex vs gender, 4 steps of sexual differentiation]
什麼啟動青春期（男女都有）？

→ **The ratio of body fat to height.**
脂肪和身高的比例。

Key: Professor 11/3 version. SRY, MIH, and TDF exist only in males, but females also go through puberty, so cross them out.
判斷關鍵：教授 11/3 版本。刪去法：SRY、MIH、TDF 只有男性有，女性也有青春期。所以初經越來越早、飲食失調可能不來。

27. Sexual dimorphism of the brain, sexual orientation （p.325–327, 334–336）

INAH-3 is larger in heterosexual men than women and smaller in gay men. Best conclusion? [Sexual dimorphism of the brain, sexual orientation]
INAH-3 異性戀男性比女性大，同性戀男性比較小。最好的結論？

→ **Some hypothalamic nuclei are sexually dimorphic, and their size is associated with sexual orientation.**
有些下視丘核是性別雙型的，大小和性傾向有關聯。

Key: "Proves" and "solely" are too strong; a correlation does not mean it is the only cause.
判斷關鍵：proves、solely 太強；相關不等於唯一原因。

28. Sexual differentiation irregularities （p.317–323）

XY, testes make testosterone, but androgen receptors don't respond (AIS). Development? [Sexual differentiation irregularities]
XY 有睪丸會分泌睪固酮，但受體沒反應。發育？

→ **No ducts of either type, a female appearance, and normal female puberty (breasts develop, infertile).**
兩套管道都沒有、外觀女性、青春期正常女性發育（有胸部、不能生育）。

Key: Testes = MIH removes the Müllerian ducts; testosterone has no effect = Wolffian ducts do not develop. Confirmed by the professor 11/3.
判斷關鍵：有睪丸＝有 MIH 殺掉 Müllerian；睪固酮沒作用＝Wolffian 長不出來。教授 11/3 確認。

XY, normal testosterone and MIH, but no 5-alpha reductase. At birth? [Sexual differentiation irregularities]
XY 睪固酮和 MIH 正常，缺 5-alpha 還原酶。出生時？

→ **Wolffian ducts, female-looking genitals, masculinized brain; masculinization begins at puberty.**
Wolffian、外觀女性、大腦男性化；青春期開始男性化。

Key: 5-alpha reductase only controls DHT for the penis: a perfect male except for the penis.
判斷關鍵：5-alpha 只管 DHT 長陰莖：perfect male except no penis。

XX fetus exposed to large amounts of testosterone. Ducts, genitals, brain? [Sexual differentiation irregularities]
XX 胎兒受到大量睪固酮。管道、外生殖器、大腦？

→ **Both duct systems, masculinized genitals, masculinized brain.**
兩套都有、男性化、男性化。

Key: Class activity version (11/3). No testes = no MIH, so Müllerian ducts still grow; testosterone works the same wherever it comes from, and genitals depend on the timing window.
判斷關鍵：課堂活動版本（11/3）。沒睪丸＝沒 MIH，Müllerian 照長；睪固酮從哪來都一樣。外生殖器要對到時間窗口。

Testosterone present but no working MIH (PMDS). Ducts? [Sexual differentiation irregularities]
有睪固酮但 MIH 沒作用。管道？

→ **Both duct systems; male genitals; masculinized brain.**
兩套都有；外生殖器男性、大腦男性化。

Key: First ask whether there are testes (MIH); then ask whether testosterone works (Wolffian).
判斷關鍵：先問有沒有睪丸（MIH），再問睪固酮有沒有作用（Wolffian）。

29. Genetic abnormalities with X and Y （p.315–316）

A tall man with small testes and low testosterone has XXY. Which is correct? [Genetic abnormalities with X and Y]
高個子、睪丸小、睪固酮低，XXY。哪個對？

→ **He is male because he has SRY; the extra X lowers his testosterone (Klinefelter syndrome).**
有 SRY 所以是男性；多一條 X 讓睪固酮偏低（Klinefelter）。

Key: The X does not outvote the Y; SRY present = testes develop.
判斷關鍵：X 不會投票贏過 Y；有 SRY 就長睪丸。

30. Hormonal and genetic effects on development, gender differences （p.320–328）

An XX girl exposed to high adrenal androgens before birth prefers rough play and scores high on mental rotation. Why? [Hormonal and genetic effects on development, gender differences]
XX 女孩出生前雄性素很高，喜歡粗野遊戲、空間能力好。為什麼？

→ **Prenatal androgens had an organizational effect that shaped her brain (CAH).**
出生前雄性素的 organizing 作用塑造了大腦（CAH）。

Key: Appears in childhood = before puberty, so it is not the activational effect.
判斷關鍵：童年就出現＝不是青春期的 activational。

31. Regulation of sex hormones in adults, parental behavior （p.337, 340, 342–343）

A bodybuilder takes high doses of testosterone; his testes shrink. Why? [Regulation of sex hormones in adults, parental behavior]
健美選手打大量睪固酮，睪丸萎縮。為什麼？

→ **Negative feedback lowers GnRH, then LH and FSH, so the testes are not stimulated.**
負回饋壓低 GnRH → LH、FSH，睪丸沒被刺激。

Key: Direction: lowered, not raised.
判斷關鍵：方向：壓低，不是調高。

Blocking vasopressin in voles' reward areas stops pair bonding; men's testosterone drops when they become fathers. Conclusion? [Regulation of sex hormones in adults, parental behavior]
擋住田鼠獎賞區的 vasopressin 就不配對；男人當爸爸睪固酮下降。結論？

→ **Vasopressin and oxytocin support pair bonding and parenting; testosterone drops when men become fathers.**
Vasopressin、oxytocin 支持配對和育兒；當爸爸睪固酮下降。

Key: Direction: drops, not rises.
判斷關鍵：方向：下降，不是上升。